



LAKE CARRIERS' ASSOCIATION



February 12, 2024

David Eng
Secretary
Federal Maritime Commission
800 North Capitol Street, N.W
Washington, D.C. 20573

Via email to secretary@fmc.gov

Subject: Request to Proceed with Petition No. P1-20, Investigation into Conditions
Created by Canadian Ballast Water Regulations in the U.S./Canada Great Lakes
Trade - Docket No. 20-10

On March 6, 2020, the Lake Carriers' Association (LCA) filed Petition No. P1-20 with the Federal Maritime Commission (FMC) alleging that conditions created by Transport Canada, an agency of the Government of Canada, in their proposed ballast water regulations (Gazette Part I) on June 8, 2019 (Volume 153, Number 23), are unfavorable to U.S.-flagged Laker shipping in the United States (U.S.)/Canada trade, pursuant to Section 19(1)(b) of the Merchant Marine Act, 1920 (Section 19) codified in 46 U.S.C. §42101. LCA requested that the FMC investigate these conditions and to adopt regulations to adjust or meet them. The LCA also provided an update of information supporting the petition on March 4, 2022.

The LCA formally requests that the FMC proceed with a ruling on Petition No. P1-20. We are making this request due to the now rapidly approaching September 8, 2024 deadline for compliance with Transport Canada's ballast water regulations, our inability to convince the Canadian Government to change their requirements regarding ballast water loading in the Great Lakes, and the recent conclusion by the U.S. EPA in a supplemental notice of proposed rule (SNPR) that ballast water management systems (BWMS) do not exist that meet the Clean Water Act's best available technology economically achievable (BAT) standard for Lakers. This letter may be placed on the public docket.

Since LCA submitted its 2022 comments, we have engaged with Transport Canada and Members of the Canadian Parliament to discuss the work that our members have undertaken to address issues with the operation of BWMS on U.S.-flagged Lakers and explain why the Transport Canada

regulation should not apply to the loading of ballast water in Canadian waters of the Great Lakes. Transport Canada has been unreceptive to these discussions has not included the U.S.-flagged Laker operators or the LCA in their industry outreach and support activities related to the regulation of ballast water. Transport Canada has also *excluded* U.S.-flagged Lakers from interactions or programs available to Canadian vessels, such as:

- Excluding impacts to the U.S.-flagged Lakers in their Regulatory Impact Analysis Assessment of the regulation,
- Excluding the LCA and its members from the October 2022 Montreal ballast water workshop,
- Excluding U.S.-flagged Lakers from participation in their Ballast Water Innovation Program (BWIP) without the use of a Canadian-based company,
- Excluding U.S.-flagged Lakers from awards of BWIP funds, even though Canadian companies received U.S. grants funding for ballast water research, and
- Excluding U.S.-flagged Lakers from applying for an exemption to the Canadian ballast water regulations or an extension of the deadline for BWMS installation similar to the Transport Canada Marine Technical Review Board review provided for Canadian vessels. In our December, 2023 visit with Transport Canada, we were told the details of a similar program would be released “soon,” which is much too late for post-2008 Lakers to install a BWMS and comply with the regulations.

Transport Canada’s inclusion of the loading of ballast water and Canada’s domestic fleet in their implementation of International Convention for the Control and Management of Ship’s Ballast Water and Sediments, 2004 (BWM Convention) was not required by their party status to the BWM Convention. To the contrary, Transport Canada’s requirements and actions taken are contrary to the BWM Convention Regulation A-4 (Exemptions). The Transport Canada ballast water regulations also do not include similar text for vessels that may be loading ballast water along the Canadian eastern or western coasts. The regulation’s specific text addressing the Great Lakes and loading of ballast water appear to have the sole purpose of disadvantaging the U.S.-flagged Laker fleet.

Transport Canada has exercised its regulatory flexibility under the BWM Convention to extend “Laker” status to Canadian-flagged vessels sailing beyond Anticosti Island (the baseline, the line of demarcation for Great Lakes pilotage waters and SOLAS requirements). Some Canadian vessels will reflag annually and trade in the Caribbean Sea when the St. Lawrence Seaway closes during the winter. Others Canadian vessel operators claim to only “occasionally sail beyond Anticosti Island.” The Canadian Government and industry are also applying political pressure on the U.S. Government to honor in U.S. regulations the special status granted by the Canadian Government to those Canadian-flagged vessels. Transport Canada is not willing, however, to grant a similar approach to U.S.-flagged vessels occasionally calling on Canadian ports to load or discharge ballast water.

In further support of LCA’s claims, the U.S. Environmental Protection Agency’s (EPA) development of the Vessel Incidental Discharge Act (VIDA) National Standards of Performance proposed on October 26, 2020 (85 FR 67818), and with a supplemental notice on October 18, 2023 (88 FR 71788), continue to exempt existing Lakers (U.S. and Canadian) from any

requirement to install a BWMS and do not align U.S. ballast water management requirements to Transport Canada's requirements. Since the EPA is not expected to release the final VIDA regulations until late September 2024, which is after Transport Canada's ballast water regulations first take effect, the LCA is basing this request on the difference between Transport Canada's final regulation requirements and the EPA's proposed rule and SNPR requirements. The LCA believes this is a reasonable approach, as the EPA's rationale included in the SNPR appears to prohibit them from moving any closer to Transport Canada's requirements in the EPA's final rule.

Canada's use of its "deemed compliance," or equipment standard, is further admission that BWMS cannot reliably meet the ballast water discharge standard in the challenging conditions of the Great Lakes, particularly for U.S.-flagged Lakers, many of which have higher ballast water volumes and flow rates than their Canadian counterparts. Transport Canada has granted waivers of its ballast water regulations for oceangoing vessels with type approved BWMS that were not able to properly operate in the Great Lakes. We are aware of one instance where a vessel that was not granted such a waiver by the U.S. Coast Guard (USCG) was granted a waiver by Transport Canada. That vessel left a U.S. port in ballast and transited to a Canadian Port to load an outbound cargo and discharge its ballast water.

The LCA has been proactive in addressing invasive species issues when needed. The Transport Canada regulation negatively impacts the operation of U.S.-flagged Lakers while providing no discernable impact on reducing invasive species in Canadian waters. Transport Canada claims that to protect Great Lakes Canadian ports from the spread of invasive species from other Canadian Great Lakes ports, the organisms must be killed twice: once prior to discharge in a U.S. port, if transported after being loaded in a Canadian port and discharged in a U.S. port, and a second time if the dead organism is loaded in a U.S. port and subsequently discharged in a Canadian port.

The following information is an update of the information the LCA previously provided to the FMC:

1. **Final Transport Canada Ballast Water Regulations: SOR/2021-120 (June 4, 2021)(Canadian Gazette, Part II, Volume 155, Number).** LCA's previous submissions did not include the final Canadian ballast water regulation that included the requirement for BWMS to be installed on post-2008 Lakers by September 8, 2024, and continued to include requirements applicable to loading ballast water in Canadian Great Lakes ports. A copy is provided in Attachment A.
2. **Updating the inventory of USCG type-approved BWMS:** Since Table 5 of LCA's July 20, 2020, comments was submitted, the USCG has type approved additional BWMS. LCA reviewed the additional USCG type approved BWMS for applicability and potential suitability to U.S.-flagged Lakers. This review did not change LCA's conclusions described in the Petition and LCA's July 20, 2020 comments to the FMC that no USCG type approved BWMS are suitable for use on U.S.-flagged Lakers. An updated table is included as Attachment B.

3. Updating the composition of the U.S.-flagged Laker fleet's pre-2009 and post-2008 vessels:

Since the 2022 LCA comments, the following changes in fleet composition have occurred:

- a. the post-2008 Laker M/V MARK W BARKER was commissioned and is in service,
- b. the pre-2009 Laker M/V JOSEPH H. THOMPSON has been removed from service and has been scrapped,

The LCA is not providing updated trade, cargo, or financial impacts in this letter, but will provide such information upon request by the FMC.

Summary

Taking into account the information presented, previously submitted comments, and interactions with Transport Canada, the LCA does not change the LCA's conclusion in Petition No. P1-20 that Transport Canada's ballast water regulations "create a 'special condition unfavorable to shipping in the foreign trade under Section 19 and under the Commission's regulations, 46 CFR Part 550.301(c) and (e)." The LCA believes the preponderance of evidence presented in our Petition in our July 20, 2020, and March 4, 2022 comments, and in this submission, demonstrate the unfavorable conditions created by Transport Canada's final ballast water regulations are designed to assure that the U.S./Canada trade is all but reserved for the Canadian-flagged fleet and is a taking of U.S. export cargoes.

Transport Canada has not released its enforcement policy regarding its ballast water regulations. We were informed that Transport Canada would release it in July 2024 for a regulation that goes into force on September 8, 2024. U.S. Laker operators are provided no process for an exemption or extension and no information regarding penalties for non-compliance until two months before this regulation goes into effect for post-2008 Lakers. They have no choice but to install and operate equipment that is not required by the U.S. Government and is unlikely to meet the discharge standard or meet the vessel's operational requirements, or they can walk away from the business, handing it to Canadian-flagged vessels. If they choose to install the equipment, they will have to remove the multimillion dollar investment when the U.S. Government determines that new technology exists that meets the BAT standard. Transport Canada is forcing U.S.-flagged vessels operators engaged in foreign trade to Canada to walk away from that market.

We request a ruling before our U.S.-flagged Lakers will be impacted by the September 8, 2024, Transport Canada compliance date.

On behalf of our 13 members operating 43 U.S.-flagged Great Lakes vessels, we thank you for the opportunity furthering our petition.

Sincerely,



James. H.I. Weakley

Attachments:

- A. Final Transport Canada Ballast Water Regulations (June 4, 2021)
- B. Updated Inventory of USCG Type Approved BWMS
- C. Updated List of U.S.-flagged Laker Fleet

cc: Chairman Daniel B. Maffei
Commissioner Rebecca F. Dye
Commissioner Louis E. Sola
Commissioner Carl W. Bentzel
Commissioner Max Vekich
General Counsel Phillip C. Hughey

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Ballast Water Regulations: SOR/2021-120

Canada Gazette, Part II, Volume 155, Number 13

Registration

SOR/2021-120 June 4, 2021

CANADA SHIPPING ACT, 2001

FISHERIES ACT

P.C. 2021-503 June 3, 2021

Whereas the annexed *Ballast Water Regulations* establish standards that are additional or complementary to the standards set out in the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, and the Administrator in Council is satisfied that those additional or complementary standards meet the objectives of the Convention;

Therefore, His Excellency the Administrator of the Government of Canada in Council, on the recommendation of the Minister of Transport and the Minister of Natural Resources with respect to the provisions of the annexed Regulations other than section 26 and on the recommendation of the Minister of Fisheries and Oceans with respect to that section 26, makes the annexed *Ballast Water Regulations* pursuant to

(a) subsections 35(1) ^a and 35.1(1) ^b, section 190 ^c and paragraphs 244(f) ^d, (g) and (h) ^e of the *Canada Shipping Act, 2001* ^f; and

(b) subsection 43(1) ^g of the *Fisheries Act* ^h.

Ballast Water Regulations

Interpretation

Definitions

1 (1) The following definitions apply in these Regulations.

Act

means the *Canada Shipping Act, 2001*. (*Loi*)

Annex

means the Annex to the Convention. (*Annexe*)

ballast water

has the same meaning as in article 1 of the Convention. (*eaux de ballast*)

ballast water exchange standard

means the standard set out in regulation D-1 of the Annex. (*norme de renouvellement des eaux de ballast*)

ballast water management

has the same meaning as in article 1 of the Convention. (*gestion des eaux de ballast*)

ballast water management system

has the same meaning as in the BWMS Code. (*système de gestion des eaux de ballast*)

ballast water performance standard

means the standard set out in regulation D-2 of the Annex. (*norme de qualité des eaux de ballast*)

BWMS Code

means the *Code for Approval of Ballast Water Management Systems*, published by the International Maritime Organization. (*Code BWMS*)

Canadian pleasure craft

means a pleasure craft that

(a) is licensed under Part 10 of the Act; or

(b) is principally maintained or operated in Canada, is not a Canadian vessel and is not registered or licensed under the laws of another state. (*embarcation de plaisance canadienne*)

Convention

means the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004, signed at London on February 13, 2004. (*Convention*)

Great Lakes Basin

means the waters of the Great Lakes, their connecting and tributary waters and the waters of the St. Lawrence River as far as the lower exit of the St. Lambert Lock at Montréal in Quebec. (*bassin des Grands Lacs*)

IBWM Certificate

means the International Ballast Water Management Certificate issued under regulation E-2 or E-3 of the Annex. (*certificat IGEB*)

Minister

means the Minister of Transport. (*ministre*)

release,

in respect of ballast water, includes leakage, pumping, pouring, emptying, dumping, spraying or placing. (*déversement*)

residual amounts

means the quantity of ballast water that cannot be removed from a ballast tank using the equipment installed on the vessel for that purpose. (*quantité résiduelle*)

sediments

means matter settled out of ballast water within a vessel. (*sédiments*)

TP 13617

means the document entitled *List of Canada's Designated Alternate Ballast Water Exchange Areas and Fresh Waters*, published by the Department of Transport. (*TP 13617*)

waters under Canadian jurisdiction

means Canadian waters and waters in the exclusive economic zone of Canada. (*eaux de compétence canadienne*)

Authorized representative

(2) For the purposes of these Regulations, a reference to the authorized representative of a pleasure craft that is not a Canadian vessel is to be read as a reference to the owner and operator of the pleasure craft.

Incorporation by reference

2 (1) Except as otherwise provided, any reference in these Regulations to a document is a reference to the document as amended from time to time.

Incorporated documents — certain terms

(2) For the purpose of interpreting a document incorporated by reference into these Regulations,

(a) “Administration” is to be read as “Minister” in respect of

(i) Canadian vessels,

(ii) Canadian pleasure craft, and

(iii) floating platforms engaged in the exploration or exploitation of the seabed and subsoil in waters under Canadian jurisdiction, including floating storage units and floating production, storage and off-loading units;

(b) “ship” is to be read as “vessel”;

(c) “survey” is to be read as “inspection”; and

(d) “discharge”, in respect of ballast water, is to be read as “release”.

Regulation A-3.5 of the Annex

(3) For the purpose of interpreting regulation A-3.5 of the Annex, the location referred to in that regulation is within 10 nautical miles from another location without encountering a physical barrier or obstruction.

Definition of *viable organisms*

(4) For the purpose of interpreting regulation D-2 of the Annex, ***viable organisms*** has the same meaning as in the BWMS Code.

Application**Application**

3 (1) Except as otherwise provided, these Regulations apply in respect of the following vessels if they are designed or constructed to carry ballast water:

(a) Canadian vessels everywhere; and

(b) vessels that are not Canadian vessels and are in waters under Canadian jurisdiction.

Activities related to oil or gas

(2) These Regulations apply in respect of vessels referred to in subsection (1) that are capable of engaging in the drilling for, or the production, conservation or processing of, oil or gas, except vessels that are on location and engaged in one of those activities in an area referred to in

(a) section 3 of the *Canada Oil and Gas Operations Act*;

- (b) subsection 8(1) of the *Canada-Newfoundland and Labrador Atlantic Accord Implementation Act*; or
- (c) subsection 8(1) of the *Canada-Nova Scotia Offshore Petroleum Resources Accord Implementation Act*.

Non-application

(3) These Regulations do not apply in respect of

- (a) vessels operated under the authority of a state that is not a party to the Convention if they are operated exclusively in the Great Lakes Basin and if they do not take on board or release ballast water in waters under Canadian jurisdiction unless it is necessary to ensure the safety of the vessel during a voyage between ports located outside Canada;
- (b) vessels that are owned or operated by a state and used only in government non-commercial service; or
- (c) vessels that carry only permanent ballast water in sealed tanks such that it is not subject to release.

Quantity

(4) For greater certainty, these Regulations apply to the management of any quantity of ballast water that may be released from a vessel.

Compliance

Authorized representative and master — Annex

4 (1) Except as otherwise provided by these Regulations, the authorized representative and the master of a vessel must ensure that the requirements of the Annex are met in respect of the vessel.

Authorized representative and master — certain provisions

(2) The authorized representative and the master of a vessel must ensure that the requirements of section 8, subsections 14(1), 15(1) and 16(1) and sections 17 to 20, 22 and 23 are met.

Master — ballast water record book

(3) The master must ensure, in respect of the ballast water record book, that

- (a) the requirements set out in the Annex to make entries are met;
- (b) the signature requirements set out in regulation B-2 of the Annex are met;
- (c) all entries are maintained on board the vessel in accordance with the requirements set out in regulation B-2 of the Annex; and
- (d) the record book is kept readily available for inspection in accordance with the requirements set out in regulation B-2 of the Annex.

Authorized representative — ballast water record book

(4) The authorized representative of a Canadian vessel or a Canadian pleasure craft must maintain the entries made in the ballast water record book, in accordance with regulation B-2 of the Annex, after the period during which the entries are to be maintained on board the vessel in accordance with that regulation.

Authorized representative and master — regulation E-1

(5) The authorized representative and the master of a Canadian vessel or a Canadian pleasure craft must ensure that the following requirements, as set out in regulation E-1 of the Annex, are met:

- (a) the reporting requirements if an accident occurs to a vessel or a defect is discovered;
- (b) the maintenance requirements; and
- (c) the requirement to obtain approval of a change that is made after an inspection.

Regulation A-4 of the Annex — Exemptions

Canadian vessels

5 (1) The Marine Technical Review Board established under section 26 of the Act is authorized, in respect of Canadian vessels, to exercise the powers of a Party to the Convention conferred by regulation A-4 of the Annex in respect of the requirements of sections 15 and 16 and of regulation B-3 of the Annex.

Other vessels

(2) On application by the authorized representative of a vessel other than a Canadian vessel, the Minister is authorized to exempt the vessel from the requirements of sections 15 and 16 and of regulation B-3 of the Annex if the application establishes that the applicable conditions described in regulation A-4 of the Annex are met.

Regulation A-5 of the Annex — Equivalent Compliance

Application

6 (1) This section applies in respect of

- (a) vessels that are less than 50 m in overall length and vessels that are not self-propelled and that are of less than 3,000 gross tonnage, if they are operated exclusively
 - (i) in waters under Canadian jurisdiction, or
 - (ii) in waters under Canadian jurisdiction and on the high seas; and
- (b) vessels that are described in regulation A-5 of the Annex.

Guidelines

(2) A vessel described in subsection (1) may, instead of complying with these Regulations, elect to comply with the requirements set out in the *Guidelines for ballast water management equivalent compliance (G3)*, published by the International Maritime Organization.

Guidelines — recommendations

(3) For the purpose of interpreting the guidelines referred to in subsection (2), with the exception of any provision relating to ballast water exchange, “should” is to be read as “must” and recommendations are mandatory.

Regulation B-1 of the Annex — Ballast Water Management Plan

Approval

7 (1) On application by the authorized representative, the Minister must approve the ballast water management plan of a Canadian vessel or a Canadian pleasure craft if the plan meets the requirements set out in regulation B-1 of the Annex.

Language of plan

(2) Despite subsection (1), the ballast water management plan must be written in English or French, or in both.

Maintenance

8 A vessel must ensure that its ballast water management plan remains up to date and reflects the means by which the vessel complies with the requirements of these Regulations.

Submission to Minister

9 (1) The authorized representative of a Canadian vessel or a Canadian pleasure craft must submit to the Minister any amendment to the ballast water management plan.

Approval of amendments

(2) The Minister must approve any amendment to the ballast water management plan if the amendment meets the requirements in respect of the plan that are set out in regulation B-1 of the Annex.

Regulation B-2 of the Annex — Ballast Water Record Book

Language of record book

10 Despite regulation B-2 of the Annex, entries made in the ballast water record book of a Canadian vessel or a Canadian pleasure craft must be written in English or French, or in both.

Regulation B-3 of the Annex — Ballast Water Management

Compliance Timeline

Certain vessels

11 (1) Despite regulation B-3 of the Annex, a vessel constructed before the day on which these Regulations come into force is not required to conduct ballast water management to meet the ballast water performance standard until September 8, 2024, and a vessel constructed before January 1, 2009 is not required to conduct ballast water management to meet that standard until September 8, 2030, if the vessel is operated exclusively

(a) in waters under Canadian jurisdiction and, if applicable, the United States' waters of the Great Lakes Basin; or

(b) in waters described in paragraph (a) and on the high seas.

Definition of *constructed*

(2) For the purposes of subsection (1), ***constructed*** has the same meaning as in regulation A-1 of the Annex.

Ballast Water Performance Standard

Deemed compliance

12 (1) A vessel using a ballast water management system to meet the ballast water performance standard is deemed to have met that standard in respect of ballast water taken on board in the Great Lakes Basin or in the eastern waters of the St. Lawrence River, if

- (a) the vessel's ballast water management system was installed before September 8, 2024 or, in the case of a vessel constructed before January 1, 2009 that is operated exclusively in waters described in paragraph 11(1)(a) or (b), the system was installed before September 8, 2030;
- (b) the vessel meets the requirements of section 8;
- (c) the vessel holds and keeps on board a valid IBWM Certificate or an equivalent document referred to in section 23;
- (d) the vessel's ballast water management system is in good working order and has been maintained and operated in accordance with the manufacturer's instructions; and
- (e) the ballast water is managed in accordance with the manufacturer's instructions for the vessel's ballast water management system, subject to any limiting operating conditions or other restrictions identified in the Type Approval Certificate evidencing the system's approval under regulation D-3 of the Annex.

Definition of *eastern waters of the St. Lawrence River*

(2) For the purposes of subsection (1), ***eastern waters of the St. Lawrence River*** means the waters of the St. Lawrence River from the lower exit of the St. Lambert Lock at Montréal in Quebec as far seaward as a straight line drawn from Cap-des-Rosiers to Pointe Ouest, Anticosti Island, and from Anticosti Island to the north shore of the St. Lawrence River along the meridian of longitude 63° W, and their connecting and tributary waters.

Alternative Methods of Ballast Water Management

Alternative methods

13 A vessel may, instead of conducting ballast water management to meet the ballast water exchange standard or the ballast water performance standard, as applicable,

- (a) conduct ballast water management in accordance with an alternative method referred to in regulation B-3.7 of the Annex if that method has been approved in accordance with the requirements of that regulation; or
- (b) release potable water, taken on board as ballast water from a public or commercial source in Canada or the United States, into waters under Canadian jurisdiction or on the high seas if it has not been mixed with other ballast water, including residual amounts, or with sediments.

Regulation B-4 of the Annex — Ballast Water Exchange

Areas for exchange

14 (1) A vessel that enters waters under Canadian jurisdiction from waters other than the United States' waters of the Great Lakes Basin and that conducts ballast water management to meet the ballast water exchange standard must conduct the exchange

- (a) in an area that is at least 200 nautical miles from the nearest land and where water depth is at least 2 000 m;
- (b) in an area described in, and in accordance with, regulation B-4.1 of the Annex, if the vessel cannot meet the requirements of paragraph (a); or

(c) in one of the alternate ballast water exchange areas designated by the Minister in TP 13617, if the vessel cannot meet the requirements of paragraph (a) or (b).

Record keeping

(2) The master of a vessel that does not comply with subsection (1) must enter the reasons in the ballast water record book.

Regulation C-1 of the Annex — Additional Measures

Releases in Canadian Fresh Waters

Exchange requirement

15 (1) In addition to meeting the requirements of the Convention, a vessel that conducts ballast water management to meet the ballast water performance standard must not release ballast water in Canadian fresh waters described in TP 13617 unless that ballast water was first exchanged in accordance with the ballast water exchange standard

(a) in an area, and in the manner, described in subsection 14(1), if the sequential method — as described in the *2017 Guidelines for ballast water exchange (G6)*, published by the International Maritime Organization — was used; or

(b) on the high seas, in any other case.

Exception

(2) Subsection (1) does not apply if the ballast water to be released was taken on board in waters under Canadian jurisdiction, in the United States' waters of the Great Lakes Basin or in waters in an area described in subsection 14(1) and has not been mixed with residual amounts taken elsewhere than from those waters.

Exception — extraordinary conditions

(3) A vessel is not required to exchange ballast water in accordance with subsection (1) if the master determines, on reasonable grounds, that the exchange would threaten the safety or stability of the vessel, its crew or its passengers because of adverse weather, vessel design or stress, equipment failure or any other extraordinary conditions.

Record keeping

(4) The master of a vessel that does not comply with subsection (1) must enter the reasons in the ballast water record book.

Saltwater Flushing

Residual amounts

16 (1) In addition to meeting the requirements of the Convention, a vessel that conducts ballast water management to meet the ballast water exchange standard must conduct a saltwater flushing of tanks that contain only residual amounts unless those residual amounts were taken on board in accordance with the ballast water exchange standard and subsection 14(1).

Saltwater flushing

(2) For the purposes of subsection (1), saltwater flushing consists of the following measures undertaken in the following order:

- (a) the addition of water to the ballast tanks in accordance with the requirements for ballast water exchange set out in subsection 14(1);
- (b) the mixing, through the motion of the vessel, of the water added under paragraph (a) with the residual amounts and any sediments that have settled in the tanks; and
- (c) the release, in accordance with the requirements for ballast water exchange set out in subsection 14(1), of the waters mixed under paragraph (b) so that the salinity of the resulting residual amounts in the tanks exceeds 30 parts per thousand or is as close as possible to 30 parts per thousand.

Regulation D-3 of the Annex — Ballast Water Management Systems

Minister's approval

17 Any ballast water management system used on a Canadian vessel or a Canadian pleasure craft must be approved by the Minister in accordance with regulation D-3 of the Annex.

Type Approval Certificate

18 A vessel must keep on board a copy of the Type Approval Certificate issued in respect of a ballast water management system installed on the vessel that evidences the system's approval under regulation D-3 of the Annex.

Regulation D-4 of the Annex — Prototype Ballast Water Treatment Technologies

Statement of Compliance

19 A vessel that is participating in one of the programs referred to in regulation D-4 of the Annex must hold and keep on board a valid Statement of Compliance referred to in the *Guidelines for approval and oversight of prototype ballast water treatment technology programmes (G10)*, published by the International Maritime Organization.

Section E of the Annex — Inspection and Certification Requirements

Certificates

IBWM Certificate

20 Every vessel to which regulation E-1 of the Annex applies must hold and keep on board a valid IBWM Certificate.

Regulation E-2 — Issuance of IBWM Certificate

Issuance of certificate

21 On application by the authorized representative of a Canadian vessel or a Canadian pleasure craft, the Minister must issue an IBWM Certificate to the vessel or pleasure craft if the requirements of an initial or renewal inspection set out in section E of the Annex are met.

Endorsements

22 A Canadian vessel or a Canadian pleasure craft that holds an IBWM Certificate must ensure that the certificate is endorsed by the Minister as required by section E of the Annex.

Vessels of Non-Parties to the Convention

Equivalent document

23 A vessel that is entitled to fly the flag of a state that is not a party to the Convention must not take on board or release ballast water in waters under Canadian jurisdiction unless that vessel holds and keeps on board a document issued by or on behalf of the government of that state that certifies that the vessel meets the requirements of the Convention.

Reporting

Canadian Ballast Water Reporting Form

24 The master of a vessel that is bound for a port, offshore terminal or anchorage area in Canada must, in the manner specified by the Minister, submit a completed Canadian Ballast Water Reporting Form to the Minister.

Consequential Amendments, Repeal and Coming into Force

Consequential Amendments

Administrative Monetary Penalties and Notices (CSA 2001) Regulations

25 Part 9 of the schedule to the *Administrative Monetary Penalties and Notices (CSA 2001) Regulations* ¹ is replaced by the following:

PART 9

Violations of the Ballast Water Regulations

Item	Column 1 Provision of the <i>Ballast Water Regulations</i>	Column 2 Range of Penalties (\$)	Column 3 Separate Violation for Each Day
1	Subsection 4(1)	1,250 to 25,000	
2	Paragraph 4(3)(a)	1,250 to 25,000	
3	Paragraph 4(3)(b)	600 to 12,000	
4	Paragraph 4(3)(c)	1,250 to 25,000	
5	Paragraph 4(3)(d)	600 to 12,000	
6	Subsection 4(4)	600 to 12,000	
7	Paragraph 4(5)(a)	600 to 12,000	
8	Paragraph 4(5)(b)	1,250 to 25,000	
9	Paragraph 4(5)(c)	1,250 to 25,000	
10	Section 8	1,250 to 25,000	X
11	Subsection 14(1)	1,250 to 25,000	X
12	Subsection 14(2)	600 to 12,000	
13	Subsection 15(1)	1,250 to 25,000	X
14	Subsection 15(4)	600 to 12,000	

15	Subsection 16(1)	1,250 to 25,000	X
16	Section 17	1,250 to 25,000	X
17	Section 18	600 to 12,000	
18	Section 19	600 to 12,000	X
19	Section 20	600 to 12,000	X
20	Section 22	600 to 12,000	X
21	Section 23	600 to 12,000	X
22	Section 24	600 to 10,000	

Aquatic Invasive Species Regulations

26 (1) Paragraph 17(1)(a) of the *Aquatic Invasive Species Regulations* ² is replaced by the following:

(a) in respect of ballast water and sediments that are carried on board a vessel designed or constructed to carry ballast water, to the persons referred to in subsection 4(1) of the *Ballast Water Regulations*; or

(2) Paragraph 17(2)(a) of the Regulations is replaced by the following:

(a) ballast water and sediments that are carried on board a vessel designed or constructed to carry ballast water; or

(3) Section 17 of the Regulations is amended by adding the following after subsection (3):

Definitions of *ballast water* and *sediments*

(4) For the purpose of subsections (1) and (2), *ballast water* and *sediments* have the same meaning as in subsection 1(1) of the *Ballast Water Regulations*.

Repeal

27 The *Ballast Water Control and Management Regulations* ³ are repealed.

Coming into Force

Registration

28 These Regulations come into force on the day on which they are registered.

REGULATORY IMPACT ANALYSIS STATEMENT

(This statement is not part of the Regulations.)

Executive summary

Issues: Ballast water, which is important for the safety and stability of vessels, can also introduce aquatic invasive species (e.g. zebra mussels) into receiving waters. In 2010, Canada acceded to the *International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004* (the Convention). Revised ballast water regulations will give effect to Canada's obligations under the Convention and further protect Canadian waters from the introduction and spread of aquatic invasive species and pathogens by Canadian and foreign vessels.

Description: These Regulations repeal Canada's *Ballast Water Control and Management Regulations* and replace them with the new *Ballast Water Regulations* (the Regulations). The Regulations apply to Canadian vessels everywhere and vessels in waters under Canadian jurisdiction. Vessels subject to the Regulations are required to develop and implement a ballast water management plan and comply with a performance standard that limits the number of organisms released. Vessels need to obtain a certificate, keep records of ballast water operations, and be subject to inspections to verify compliance. Most vessels will need to install a Ballast Water Management System (BWMS) to comply with the Regulations. Smaller vessels and certain non-self-propelled vessels have the option of an equivalent compliance regime more tailored to their operations and size of their vessels. The Regulations take into account differences between the United States regime and the Convention, providing sufficient time for vessel owners to install BWMS for use in the North American market and the challenging water quality conditions on the Great Lakes and St. Lawrence River.

Rationale: The Regulations give effect to Canada's international obligations under the Convention, which entered into force in 2017, and further reduce the environmental risks posed by ballast water release. The implementation of the Convention's performance standard by foreign and domestic vessels will reduce the environmental and economic impacts of invasive species introduced to Canadian waters. As a result, the Regulations contribute directly to a clean and healthy environment, align with the need for a strong and mutually beneficial North American partnership, and support a prosperous Canada through global commerce. The Regulations impose a present value total cost of \$280.47 million. Private vessel owners will incur the majority of the costs associated with the Regulations. The present value total benefits will be \$981.85 million, resulting in a net benefit of \$701.38 million.

Issues

Through these Regulations, Canada is giving effect to the international obligations it took on in acceding to the Convention in 2010. Canada is also regulating domestic vessels, which were exempt from the former regulations despite posing economic and environmental risks associated with the spread of invasive species throughout Canadian waters (and in inland waters shared with the United States).

Background

Ballast water is water taken on board a vessel to control trim, list, draught, stability or stresses of the vessel. For example, it may be taken up or released when cargo is unloaded or loaded, or when a vessel needs extra stability in poor weather. Ballast water carried on board a vessel is also a vector by which harmful aquatic organisms and pathogens are unintentionally transported around the world. Invasive species have been implicated in vast reductions in, or outright extinction of, indigenous fish and the devastation of local fisheries. Invasive species also harm plants and animals, facilitate algal blooms, degrade beaches, impair fisheries, disrupt infrastructure, lower property values, and create prevention and control expenses.

Canada first introduced voluntary guidelines to address the introduction of invasive species to the Great Lakes by international shipping in 1989. In June 2006, mandatory national rules were introduced pursuant to the former *Canada Shipping Act*: the *Ballast Water Control and Management Regulations*. There have been no substantive policy changes to the 2006 regime since its introduction. The Regulations were updated on October 27, 2011, to bring them under the regime

of the new *Canada Shipping Act, 2001*, which had entered into force in the interim. On February 22, 2017, the Regulations received minor amendments to address issues that were raised by the Standing Joint Committee for the Scrutiny of Regulations to correct discrepancies between the English and French versions.

As shipping is a global industry and the movement of aquatic invasive species is a global issue, a global approach is needed for its regulation. In 1989, Canada initiated the discussion concerning invasive species in ballast water at the International Maritime Organization (IMO). Canada then actively participated in the development of the Convention, which was adopted in 2004 in order to prevent, minimize, and ultimately eliminate the risks to the environment, human health, property, and resources arising from the transfer of harmful aquatic organisms and pathogens. In 2010, Canada acceded to the Convention, which entered into force on September 8, 2017.

In 2014, a peer-reviewed national risk assessment, prepared by Fisheries and Oceans Canada (DFO) for Transport Canada (TC), found that all ballast water movements represent a risk for introducing and spreading aquatic invasive species in Canada. This study found that the international vessels travelling to the Great Lakes from overseas represented the lowest risk category of vessels, due to full compliance with requirements that they replace ballast water contents with water taken up from the open ocean. By comparison, international vessels visiting Canada's Atlantic and Pacific coasts posed a high risk; international vessels visiting the Arctic posed an intermediate risk; and domestic Great Lakes vessels posed a higher risk of spreading invasive species to new areas, exacerbating associated negative effects.

Objective

The objectives of the Regulations are to

- reduce the risk to Canada's environment and economy associated with the introduction and spread of aquatic invasive species released through the ballast water of foreign and domestic vessels;
- give effect to the Convention in Canada and address Canada's international obligations, including the protection of foreign environments;
- facilitate adoption by ships of nascent ballast water treatment systems, especially in the technically challenging Great Lakes–St. Lawrence River region and in environmentally sensitive Canadian fresh waters; and
- maximize compatibility with the differing and evolving United States ballast water regime.

Description

These Regulations repeal the former *Ballast Water Control and Management Regulations* and replace them with a new set of regulations, the *Ballast Water Regulations*. These Regulations

- apply to Canadian vessels everywhere and all vessels in waters under Canadian jurisdiction;
- incorporate by reference the Convention's requirements;
- impose requirements based on the vessel's length, its ballast water capacity, its date of construction, and its area of operation; and
- maintain foundational requirements from the former regulations that can still be applied to the amended regime, such as reporting requirements.

Vessels to which the Regulations apply are divided into three groups. ⁴

1. *International vessels*

Vessels that operate internationally will be required to be in compliance with the Convention regime, which requires that vessels

- have on board and implement an approved vessel-specific Ballast Water Management Plan;
- be surveyed and carry a Ballast Water Management Certificate;
- meet a performance standard that limits the number of organisms capable of reproducing in order to reduce the risk of aquatic species invasions (vessels are expected to use a BWMS to meet the performance standard);
- record ballast water operations and maintain a Ballast Water Record Book on board; and
- be subject to inspections in ports or offshore terminals to ensure compliance.

These vessels will also be subject to some former provisions that remain relevant and are not part of the Convention regime

- to flush otherwise-empty ballast tanks with open ocean water in order to reduce the risk posed by any residual ballast water and sediments;
- to exchange and flush ballast tanks in addition to meeting the performance standard when travelling to Canadian fresh waters (from outside of waters under Canadian jurisdiction, the Great Lakes and the high seas);
- to conduct any exchange or flushing operation in waters at least 2 000 metres deep, whenever possible; and
- to report on the provenance and management of ballast water released in Canada.

The Regulations will require all vessels travelling internationally to comply with the Convention's requirements. The Convention requires vessels travelling internationally and built on or after September 8, 2017, to meet the performance standard when the vessel is launched. Conversely, as per the Convention, vessels built before September 8, 2017, will be required to meet the performance standard using a phased-in approach from 2019 to 2024. During this phased-in approach and prior to being required to comply with the performance standards, these vessels are required to continue to exchange their ballast water in mid-ocean in order to reduce the number and viability of organisms released. The requirement for these vessels bound for fresh waters to exchange ballast water mid-ocean or in an alternative ballast water exchange zone, will be reviewed by TC by 2030 based on scientific evidence for its environmental benefit and on its effect on vessel operations.

2. *Domestic and Great Lakes vessels*

These vessels include those that operate exclusively in waters under Canadian jurisdiction, as well as those that operate there and at United States Great Lakes ports and/or on the high seas. To address the spread of species within Canada, domestic and Great Lakes vessels will be required to comply with the same applicable requirements as vessels in Group 1 above. However, those vessels constructed in or after 2009 will have until September 8, 2024, to come into compliance with the performance standard, while those vessels constructed before 2009 will have until September 8, 2030, to come into compliance. Non-party vessels (e.g. United States vessels) that transit through Canadian waters of the Great Lakes Basin without loading or unloading ballast water (other than ballast water necessary for the purpose of ensuring the safety of the vessel on a voyage between non-Canadian ports) will be exempt from the Regulations (see Regulatory Development below).

3. Vessels of non-parties

The Convention requires Canada to apply the requirements of the Convention to vessels of non-parties to ensure that no favourable treatment is given to such vessels. The Convention's requirements include the development of approved ballast water management plans for meeting the Convention's performance standard wherever ballast water is discharged — even if the ballast is ultimately discharged into waters of non-parties. The Regulations therefore require that vessels that load or discharge ballast water in Canada hold and keep on board a document of compliance issued by, or on behalf of, their flag state that certifies that the vessel meets the requirements of the Convention.

4. Vessels subject to the equivalent compliance regime

The Convention allows Canada to establish equivalent compliance requirements for certain international pleasure craft, and search and rescue craft that carry less than eight cubic metres of ballast water and are less than 50 metres in length. The Regulations will do so for these vessels by giving effect to the IMO guidelines for equivalent compliance. For reasons of practicality and feasibility, the Regulations will also allow vessels less than 50 metres in length, as well as non-self-propelled vessels with a gross tonnage of less than 3 000 tons, to follow the equivalent compliance regime if they operate exclusively in waters under Canadian jurisdiction, or in those waters and on the high seas. Equivalent compliance refers to a set of methods and best practices approved by the IMO that allows vessel owners to determine how best to manage ballast water on board their vessel, as installing and operating BWMS and meeting all of the requirements under the Regulations is not always feasible.

5. Exemptions

The Convention provides a mechanism for internationally coordinated time-limited, renewable, risk-based exemptions under its regulation A-4. TC will consider applications for exemption on a case-by-case basis from vessel owners that meet exemption requirements under regulation A-4. However, because unmanaged ballast water discharges have been found to pose a substantial risk in all areas of Canada, TC does not anticipate that Canada will grant a significant number of exemptions. TC will develop guidelines for applicants and applications will be reviewed in conjunction with DFO.

In accordance with the Convention, the Regulations do not require management of ballast water loaded and discharged at the same location, providing that no mixing with unmanaged ballast water has occurred. The Convention does not define the term “same location.” The Regulations define “same location” to mean locations within 10 nautical miles, without crossing an obstruction or barrier. This aligns with the United States Environmental Protection Agency's definition of “short-distance voyages,” and encompasses short-distance ferry routes, as well as the operation of support vessels within many Canadian ports.

Regulatory development

Consultation

TC has consistently engaged stakeholders concerning ballast water through public meetings of the Canadian Marine Advisory Council (CMAC) since before Canada's 2010 accession to the Convention. In addition, the Regulations reflect years of dialogue with industry, scientists, engineers, United States legislators and regulators, and international partners, notably with

- Canadian and United States vessel owners, business associations, scientists, legal associations, landowner associations, United States jurisdictions, and BWMS manufacturers who submitted public comments in March 2013

in response to an October 2012 TC discussion paper on a proposed regulatory approach to give effect to the Convention;

- bilateral discussions with the United States and its regulators in accordance with the 2012 Great Lakes Water Quality Agreement;
- vessel owners and academics who participated as scientific peer reviewers for a national risk assessment on ballast water and aquatic invasive species in March and June 2013, as well as for a scientific assessment of ballast water exchange plus treatment in February 2018;
- expert engineering consultants who undertook studies commissioned by TC in 2013 on the availability of BWMS for the Great Lakes, and whose subsequent technical dialogue with Canadian and United States vessel owners was published by TC in 2015; and
- members of a Government-Industry Working Group on Ballast Water formed in early 2017, which included representatives of Canadian owners of domestic and international vessels, and which held eight in-person meetings culminating in an agreement in principle to the regulatory approach.

In addition to these consultations and processes, Canada has engaged at dozens of IMO meetings and held many informal and bilateral discussions with Canadian and United States vessel owners.

Prepublication in the *Canada Gazette*, Part I

The Regulations were prepublished in the *Canada Gazette*, Part I, on June 8, 2019, providing a 90-day comment period. A total of 18 different stakeholder groups consisting of marine industry, environmental organizations, and other government departments submitted comments.

The majority of comments were in support of the proposed Regulations and in favour of Canada's efforts to take a leadership role in advancing environmental protections. Many environmental stakeholders, other government departments and agencies, and industry groups expressed support for the Regulations and for TC's approach to implement the Convention and accede to Canada's international commitments.

Some industry stakeholders expressed concerns in a number of areas. The comments ranged from minor changes to text to clarifications on more complex issues surrounding feasibility, costs, effectiveness, implementation, and compatibility with the United States regime.

As many of the stakeholders shared similar concerns, below are the overarching issues grouped together by theme and summarized.

Cost-benefit analysis

Stakeholder comments on the cost-benefit analysis were mainly on the accuracy of estimated costs and benefits, and the exclusions of non-self-propelled barges and United States flagged vessels.

Underestimation of costs

Four stakeholders were of the view that the installation cost of BWMS was underestimated, as the low-end estimate of the cost of installing BWMS makes the assumption that shipowners will choose the least-cost option, which may not be the case as this decision would depend on multiple factors, and that those factors could drive the costs higher than the estimated cost of \$632.39 million presented in the Regulatory Impact Analysis Statement in the *Canada Gazette*, Part I.

TC maintains that using the least-cost option is a common practice in the cost-benefit analysis that follows the fundamental assumption in economics that firms seek to maximize their profits. Furthermore, the cost-benefit analysis assessed impacts as a result of the proposed Regulations, which only require that a vessel meet the performance standard. For some of the impacted vessels, this requirement entails the installation of a BWMS. The decision to install a more expensive system is a business decision, not a regulatory requirement. Therefore, such business decisions were not considered in the cost-benefit analysis. However, a sensitivity analysis was conducted as part of the cost-benefit analysis to address uncertainties around the installation cost of BWMS and presented results under different scenarios.

Overestimation of benefits

One of the comments expressed concerns that the estimated benefits of \$1,296.11 million in the Regulatory Impact Analysis Statement, which was published in the *Canada Gazette*, Part I, were too high based on the assumption that 15% of non-native aquatic species introduced to Canada would cause significant environmental and economic damage. This concern was based on the view that the United States Congressional Office of Technology Assessment's (OTA) Report on Harmful Nonindigenous Species (1998) [PDF] was not specific to the marine context.

TC notes that the assumptions on the overall number of prevented non-native species were based on a 2019 DFO study,⁵ as well as the United States Congressional Office of Technology Assessment's report. The 15% figure also aligns with the United States Coast Guard regulatory analysis.⁶ In the sensitivity analysis presented in this document, different values on the percentage of non-native aquatic species causing severe impact were considered, both lower (10%) and higher (18%) than the central case (15%).

Exclusion of non-self-propelled vessels under 3 000 gross tons

Two stakeholders commented that the costs associated with fitting barges with BWMS were not included in the cost-benefit analysis, while other comments suggested that this subset of vessel was not properly analyzed.

TC confirms that barges and their associated costs were included in the cost-benefit analysis presented in the Regulatory Impact Analysis Statement, which was published in the *Canada Gazette*, Part I. In that analysis, it was also acknowledged that there was a lack of information about the various types of technical arrangements (e.g. piping, electrical systems, etc.) on board these vessels, and, therefore, the installation of the Hyde Guardian BWMS on the Ranger III was used as a proxy to estimate costs for these vessels.

In light of stakeholder concerns expressed during the prepublication (*Canada Gazette*, Part I) comment period, and recognizing the challenges associated with fitting non-self-propelled barges with BWMS, the Regulations have been amended to permit non-self-propelled vessels under 3 000 gross tonnage to comply with the Regulations through the equivalent compliance regime. The cost-benefit analysis has been revised accordingly.

Exclusion of non-Canadian flagged vessels

Some stakeholders highlighted that the cost-benefit analysis did not include the cost to non-Canadian flagged vessels, in particular United States flagged vessels operating in the Great Lakes–St. Lawrence River.

As per the Treasury Board Secretariat's Cabinet Directive on Regulation and the Policy on Cost-Benefit Analysis, costs and benefits that are in scope are those that are attributed to “Canadians” defined as the Canadian community as a whole. Therefore, regulatory impacts on international shipowners, including United States flagged vessels operating in the Great Lakes–St. Lawrence River, are outside the scope of the cost-benefit analysis for the Regulations.

Competitiveness impact analysis

One of the stakeholders expressed their concern that the cost-benefit analysis of the proposed Regulations did not estimate the impacts on Canadian manufacturing sectors, in particular, the impacts to the steel sector. The stakeholder highlighted that the steel sector relies heavily on international, Canadian and United States vessel operators for their input materials and product distribution. They expressed concerns that the proposed Regulations would result in a shortage of vessel capacity and increase transportation rates, which would consequently affect their global competitiveness.

As per Treasury Board Secretariat's *Cabinet Directive on Regulation* and the *Policy on Cost-Benefit Analysis*, costs and benefits that are in scope are those involving the direct impact on the regulated community, i.e. Canadian vessel owners who would be required to install a ballast water treatment system. While impacts on Canada's manufacturing sectors are important, they are considered indirect impacts and therefore are not analyzed in the cost-benefit analysis.

TC notes that the Regulations do not limit the number of vessels that may operate in Canada, and are needed to address environmental risks posed to Canadian waters, and to meet Canada's international obligations. Approximately 91% of the global fleet is subject to the Convention, which limits impacts on manufacturing competitiveness. Interactions with the United States regime are further discussed below.

Compliance timeline

Comments pertaining to compliance timelines focused on the timeline for ships that operate only in waters under Canadian jurisdiction, or if applicable, in the United States waters of the Great Lakes and the high seas. Two commenters raised feasibility concerns with regard to the compliance timelines, while other commenters stressed the need for urgent action and faster implementation in the interests of environmental protection.

TC has taken into consideration the logistical challenges involved in sourcing and outfitting a large number of Great Lakes ships with BWMS in this region in a short timeframe, taking into account matters, such as the passage of time since the Regulations were proposed; societal challenges associated with the COVID-19 pandemic; the challenges reported by industry in finding sufficient installation facilities and personnel to physically install BWMS; the additional time needed to address technical challenges faced by older Great Lakes ships that were designed prior to the era of ballast water management; and the evolving United States regime.

In light of these factors, the Regulations have been updated to extend the compliance timeline for vessels built before 2009. These vessels will now be required to comply with the performance standard by September 8, 2030 (as opposed to 2024), if they remain in waters under Canadian jurisdiction, the United States waters of the Great Lakes, and the high seas. This timeline, which was informed by DFO risk assessments (DFO 2020), provides additional time to address the technical and logistical challenges conveyed by both Canadian and United States shipowners to TC through public comments in the interests of a feasible transition. This timeline also takes into account the United States regulatory calendar established by the United States *Vessel Incidental Discharge Act*, which requires that the United States Environmental Protection Agency review its standards in 2025 and 2030, and positions Canada and the United States to continue work in partnership with shipowners to overcome remaining BWMS installation challenges (especially for the older vessels), while ensuring progress for the environment and towards addressing Canada's international obligations.

Deemed compliance

The Regulations provide that a vessel using a ballast water management system to meet the ballast water performance standard is deemed to have met that standard in respect of ballast water taken on board in the Great Lakes Basin or in the eastern waters of the St. Lawrence River under certain conditions.

TC included this “deemed compliance” provision to address the risk that BWMS might not consistently meet the performance standard in the Great Lakes region due to the challenging water quality conditions in this area (e.g. very silted, cold, fresh or brackish water) in order to allow the Convention to be feasibly implemented in this region and increase environmental protection as soon as possible.

Stakeholders expressed concern that BWMS might not sufficiently protect the environment if standards are not consistently met, and that this might limit the benefit of installing a BWMS. However, despite performance concerns, the use of BWMS would substantially reduce the environmental and economic impacts associated with the spread of non-native species by Great Lakes ships. DFO's national ballast water sampling demonstrates that ships using BWMS, including on the Great Lakes, are consistently meeting the D-2 standard for smaller organisms like phytoplankton (100% of 39 tests). For larger zooplankton organisms, these ships meet the standard roughly 50% of the time (based on 37 tests) and otherwise limit the number of zooplankton released to a small fraction of the typical number found in unmanaged Great Lakes ballast water. A median of 43 zooplankton per cubic metre was released when the standard of 10 zooplankton was not met. By comparison, a median of around 53 000 zooplankton per cubic metre was found during sampling of 87 unmanaged releases of ballast water from Great Lakes ships (DFO 2014). This demonstrates the effectiveness of current levels of BWMS performance for reducing the risk of spreading invasive species on the Great Lakes. To be conservative, the cost-benefit analysis assumes this level of BWMS performance; however, TC expects performance using BWMS to improve with additional training and experience.

The date by which a BWMS must be installed to benefit from deemed compliance has been extended to align with the amended compliance timeline in order to reflect the challenge faced by the older vessels on the Great Lakes that were designed and constructed before the era of ballast water management. Two stakeholders raised concerns that ships should not be able to rely on deeming indefinitely but should instead be required to meet the performance standard at some point in the future. TC considered these concerns in the context of the 2020 advice from DFO concerning the substantial reductions in establishments expected using BWMS at current performance levels. Given the substantial cost associated with the installation of a BWMS, the Regulations deem compliance with the performance standard for the life of the installed BWMS as long as the conditions set out in the Regulations are met.

One commenter suggested amending the deeming provision to require the use of a BWMS only to the extent practicable while maintaining normal ballast water operations. TC considered the proposal, but concluded that doing so would not provide desired environmental benefits as it could reduce the amount of ballast water that would be treated and it would be difficult to enforce. TC also considered requests for greater clarity on acceptable contingency measures in the event of BWMS failures. These types of matters will be addressed by TC, as needed, through guidance materials such as Ship Safety Bulletins and Technical publications.

Ballast water exchange

Combining exchange with the use of a BWMS

The proposed Regulations set out a requirement that vessels operating in Canadian fresh water after operating outside the Exclusive Economic Zone exchange ballast water until September 8, 2024, as well as meet the performance standard once required to do so (“exchange plus treatment”). One stakeholder raised the view that ships should not be required to

conduct exchange plus treatment as long as they use a BWMS certified to work in fresh water, as this increases regulatory complexity and poses operational challenges. Other stakeholders were of the view that the measure should only be applied to the Great Lakes region (to align with the United States), or that the measure should be expanded to include all waters regardless of salinity. One stakeholder supported the provision as a transitional measure until BWMS performance improves, while two other stakeholders felt it should be an indefinite measure.

Given the breadth of views received, TC consulted with and received additional science advice from DFO in 2020 on the locations and timeframe for exchange plus treatment requirements. As a result of those consultations, the following decisions were made:

- **Locations:** The Regulations will require exchange plus treatment for Canadian fresh waters as originally proposed. This will ensure that environmental benefits accrue to all Canadian fresh waters — notably those around Montréal and the city of Quebec (which receive large volumes of ballast water), while other Canadian fresh waters receive proportionate benefits.
- **Timeframe:** The Regulations have been amended to require the use of exchange plus treatment on an indefinite basis. The practice will provide ongoing environmental benefits — even if ballast water would otherwise meet the performance standard — which also aligns with the indefinite approach to exchange plus treatment in the United States.

While the evidence does not presently support limiting the time horizon for exchange plus treatment or limiting the fresh waters in which the measure is required as requested by many stakeholders, TC recognizes that relevant information may emerge in the future (e.g. on environmental protection, on BWMS performance and reliability, and on implications for vessel operations and equipment). Therefore, TC intends to review the exchange plus treatment provision by 2030 and consider additional actions or changes as needed. This will permit time to build experience with the provision and ballast water management in general, to assess improvements in BWMS reliability, to undertake any further scientific field work and/or risk assessments that may be necessary, and to continue discussions with the United States and international partners.

Location and depth of ballast water exchange

While the *Ballast Water Control and Management Regulations* required that ballast water exchange be undertaken in waters that are at least 2 000 metres deep, the proposed Regulations aligned with the Convention requirement for 200-metre depths. Some comments expressed concern that the proposed approach would increase risk to Canadian ecosystems close to shore, where depths are lower. In light of the comments, TC conducted additional consultations with DFO and the Regulations have been amended to stipulate that vessels conducting ballast water exchange are required to do so in waters at least 2 000 metres deep, whenever possible. Where a vessel cannot comply with the 2 000-metre depth requirement, it would be required to meet the 200-metre depth in accordance with the Convention standard, and the Master of the vessel would be required to enter the reasons in the ballast water record book.

One comment also requested that alternate ballast water exchange areas generally be located in deep waters, far from shore, and outside Marine Protected Areas and other sensitive coastal areas. Alternate ballast water exchange areas, which are based on scientific advice from DFO, are set out in standard TP 13617. Since the prepublication of the proposed Regulations, this standard has been amended to reflect the most recent specific science advice concerning Arctic zones. Amendments to the standard are undertaken when warranted and in accordance with scientific advice provided by DFO.

Ballast water exchange during equivalent compliance

Three comments noted that ships operating in the equivalent compliance regime may not always be in a position to undertake mandatory ballast water exchanges. In light of these concerns, the Regulations have been amended to clarify that vessels in the equivalent compliance regime should exchange ballast water to the extent practicable. TC will monitor compliance with this requirement and take corrective actions as necessary.

Requirements of the United States

Several stakeholders raised concerns that since Canada is aiming to align the Regulations with the Convention, and that the United States is not a signatory to the Convention, there could be areas where the two regulatory regimes do not align. This has been noted as a particular issue for Canadian vessels operating on the Great Lakes and St. Lawrence River that trade between Canada and the United States.

The Convention, which was adopted in 2004, and entered into force in 2017, applies to the Great Lakes. The Regulations are needed to fulfill Canada's international obligations and to reduce the introduction and spread of aquatic invasive species through vessel ballast water.

The United States ballast water regime is under review and will continue to evolve in the coming years in accordance with its *Vessel Incidental Discharge Act*. TC will also continue to engage with the United States (e.g. under the Great Lakes Water Quality Agreement) to promote a compatible approach that addresses the environmental and economic risks of invasive species, and that minimizes uncertainty for the regulated community so that stakeholders in both countries are able to comply with both regimes. TC will continue to monitor developments in the United States and will consider additional actions as appropriate.

Application to Great Lakes vessels of the United States

Commenters generally supported the approach to regulating United States vessels that operate in Canadian ports, noting the contribution of Great Lakes shipping to the spread of invasive species. However, two commenters felt it inappropriate for Canada to regulate vessels that load ballast water in Canada and release it in the United States, suggesting trade and competitiveness concerns, a lack of environmental benefit to Canada, disproportionate cost impacts on United States vessels, and a lack of basis under the Convention.

After considering these concerns, TC has determined that the approach proposed in the draft Regulations is appropriate because the approach

- will address environmental risks associated with the movement of non-native species in unmanaged Great Lakes ballast water; ²
- is fair in setting the same rules for Canadian and United States Great Lakes ships, while cost differences that arise from commercial decisions (e.g. vessel design and capacity) are outside Canada's control; and
- closely follows the Convention in requiring vessels to implement an approved ballast water management plan to meet the standards regardless of the location of discharge.

However, TC also considered these commenter concerns in light of the technical challenges associated with the transition to ballast water management on older Canadian and United States Great Lake vessels. These concerns informed the decision to provide more time for compliance with the standards. Vessels built before 2009 will now have until September 8, 2030, to comply (as opposed to 2024).

TC proposed not to regulate United States Great Lakes vessels in transit between United States Great Lakes ports (described as vessels that do not take on or release ballast water in Canada). The Regulations have been adjusted to address comments requesting that transiting vessels be permitted to take on or release ballast water in Canada for the purpose of ensuring safety on a voyage between non-Canadian ports.

Potable water as an alternative compliance method

Two commenters suggested that the discharge of potable water, used as ballast water, be allowed, in order to facilitate compliance by smaller vessels or those that infrequently release ballast water.

This issue remains before the IMO, and TC is following the development of the discussions there. In the case of vessels discharging ballast water into waters under Canadian jurisdiction or on the high seas, the Regulations were amended to allow for the use of potable water as ballast, if drawn from a municipal or commercial source in Canada or the United States, and not mixed with other ballast water (including residual amounts) or sediments.

Comments provided following prepublication

TC has continued to engage with stakeholders and United States partners since prepublication to address outstanding questions and prepare for implementation.

TC received follow-up letters from stakeholder groups in 2020 and 2021, which reiterated concerns expressed during the prepublication consultation period. These letters included concerns about (i) the costs being underestimated and the benefits being overestimated; (ii) the differing approaches of Canada and the United States to ballast water management in the shared waters of the St. Lawrence and Great Lakes; and (iii) compliance dates in light of technical challenges for vessels operating in these waters.

The above "Prepublication in the *Canada Gazette*, Part I" section explains how TC has addressed these concerns.

COVID-19 pandemic

Since the onset of the COVID-19 pandemic in March 2020, TC has been able to engage virtually with stakeholders through the national CMAC meeting held in November 2020, and also held several virtual meetings with industry stakeholders in December 2020 and January 2021 to provide updates and discuss concerns.

As previously noted, the Regulations provide lead time for stakeholders to come into compliance with the performance standard. The Regulations apply to vessels depending on their operations, ballast water capacity and size, and the compliance dates have been staggered accordingly. As noted above, partly due to COVID-19, the compliance timelines for older vessels have been extended in the final regulations. Vessels constructed in or after 2009 will have until September 8, 2024, to come into compliance with the performance standard, while those vessels constructed before 2009 will have until September 8, 2030. It is expected that, despite the ongoing pandemic, stakeholders will be able to meet these compliance dates.

Modern treaty obligations and Indigenous engagement and consultation

In accordance with the *Cabinet Directive on the Federal Approach to Modern Treaty Implementation*, an analysis was undertaken to determine whether the Regulations are likely to give rise to modern treaty obligations. This assessment examined the geographical scope and subject matter of the Regulations in relation to modern treaties in effect.

Treaty provisions are identified in the Nunavut Land Claim Agreement, the Nunavik Inuit Land Claims Agreement, and the Eeyou Marine Region Land Claims Agreement, which obligate the federal government to consider advice and recommendations regarding marine areas.

TC has responded to interest from Indigenous groups and rights holders concerning ballast water over an extended period of time. This includes an extended dialogue with the Nunavut Impact Review Board in 2012 during hearings on the Mary River Mine Project, including several in-person appearances by TC officials, and briefings given to the Chiefs of Ontario by TC's Ontario Region in the context of the Canada-Ontario Agreement.

TC has also consulted on the Regulations through the regional CMAC, which includes representatives of Indigenous groups. In terms of both environmental benefits and impact on vessel owners, the Regulations are expected to affect Indigenous peoples in the same ways as non-Indigenous Canadians.

Instrument choice

For these Regulations, no other options were considered as its intent is to align Canadian regulations with the *International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004*, to which Canada is a party.

Regulatory analysis

Benefits and costs

The Regulations will require that vessels comply with the Convention's performance standards, resulting in incremental one-time capital costs and ongoing operational costs to vessel owners, as well as compliance promotion and enforcement costs to the Government of Canada. The present value total costs of the Regulations are expected to be \$280.47 million between 2021 and 2043, with capital and operational costs to private vessel owners representing the majority of the costs.

The requirements set out in the Regulations will reduce the risk of introduction and spread of invasive species in Canadian waters while also addressing Canada's international obligations under the Convention. Based on advice from DFO, and the mitigation costs of controlling zebra mussel invasion in the Ontario Great Lakes area, the present value benefits were estimated to be \$981.55 million between 2021 and 2043. As a result, the net benefit is \$701.38 million.

The information provided in this section is a summary of the overall findings of the analysis. A comprehensive cost-benefit analysis report is available upon request.

Analytical framework

Benefits and costs associated with the Regulations are assessed based on comparing the baseline scenario against the regulatory scenario, in accordance with the Treasury Board Secretariat's *Policy on Cost-Benefit Analysis*. The baseline scenario depicts what is likely to happen in the future if the Government of Canada does not implement the Regulations. The regulatory scenario provides information on the intended outcomes as a result of the Regulations.

This analysis estimates the impact of the Regulations over a 23-year analytical period from 2021 to 2043, with the year 2021 being when the Regulations are registered. The 23-year timeframe is chosen to take into consideration various compliance dates and the expected useful life of a BWMS (20 years).

Unless otherwise stated, all total values are expressed in 2017 Canadian dollars, discounted to 2021 at a 7% discount rate. All unit values are in 2017 Canadian dollars, undiscounted.

Affected stakeholders

The Regulations will apply to Canadian vessels everywhere, and foreign vessels operating in waters under Canadian jurisdiction. Requirements for impacted vessels are based on their length, built year, ballast water capacity, and the area where the vessel operates.

In the baseline scenario, vessels that operate exclusively in Canadian waters or exclusively in the Canada-United States Great Lakes and St. Lawrence River would not be required to manage their ballast water. Vessels operating internationally (excluding those operating exclusively in the Canada-United States Great Lakes) would be required to exchange their ballast water prior to releasing in Canadian waters.

In the regulatory scenario, the requirements vary based on the area where vessels release their ballast water, and the vessel characteristics. For the purpose of the cost-benefit analysis, in-scope Canadian vessels are grouped into four categories. Table 1 summarizes requirements, and new actions resulting from the Regulations.

1. Canadian vessels operating internationally, releasing ballast water in Canadian fresh water ports (excluding those operating exclusively in the Canada-United States Great Lakes)

While these vessels will continue to exchange their ballast water, the Regulations will require that they install a BWMS and treat their ballast water as well. These vessels are expected to comply with the performance standards by September 8, 2024. ⁸

2. Canadian vessels operating internationally (including in the United States), releasing ballast water in Canadian coastal waters

These vessels will no longer be required to exchange their ballast water. They will instead be required to install a BWMS, and treat prior to release. These vessels are expected to comply with the performance standards by September 8, 2024.

3. Vessels operating exclusively in the United States and Canada Great Lakes and St. Lawrence River

Vessels that operate exclusively in the United States-Canada Great Lakes and St. Lawrence River will be required to install a BWMS, and treat ballast water prior to release in Canadian waters. The date by which they will have to comply depends on the year in which the vessel was built. Vessels built in 2009 or later will have to comply with the Regulations by September 8, 2024, while those built prior to 2009 will have until September 8, 2030, to comply. Table 1 below lists the number of vessels by their area of operations and required compliance dates.

4. Canadian vessels operating exclusively in Canadian waters

The Regulations will require certain vessels under this category to install a BWMS to meet performance standards, while others will be subject to a best-effort equivalent compliance regime.

a. Vessels operating exclusively in Canadian waters that are either:

- i. Self-propelled AND longer than or equal to 50 metres in length; OR
- ii. Non-self-propelled AND more than or equal to 3 000 gross tonnage

These vessels will be required to meet performance standards that require installing a BWMS and treat their ballast water prior to release. The compliance date for this group will depend on the year in which the vessel was built, following the same timeline as those operating exclusively in the United States and Canada Great Lakes and St. Lawrence River (see

Table 1 for details).

b. Vessels operating exclusively in Canadian waters that are either:

- i. Self-propelled AND less than 50 metres in length; OR
- ii. Non-self-propelled AND less than 3 000 gross tonnage

This group of vessels will be permitted to comply with either the equivalent compliance regime, or the performance standards (installing and operating a BWMS). For the purpose of this analysis, all vessels under this category are assumed to elect the equivalent compliance regime as their compliance method.

The equivalent compliance regime comes into force the day upon which the Regulations are registered; however, enforcement actions would not be invoked until after the Regulations are published in the *Canada Gazette*, Part II. To the extent possible, owners will be required to manage their vessel's ballast water in accordance with Convention guidelines.

Table 1 shows the number of affected vessels by compliance dates, as well as comparisons between the baseline and regulatory requirements.

Table 1: Number of affected Canadian vessels, by requirements and new actions

Vessel category	Affected vessels	2024 compliance	2030 compliance	Baseline requirements	Regulatory requirements	New actions
Vessels operating internationally and releasing in Canadian fresh water ports	15	15	0	Exchanging ballast water	Exchanging plus treating ballast water prior to release in Canadian waters	Installation of BWMS, and ongoing operation of system (treatment). They will continue to exchange their ballast water.
Vessels operating internationally releasing in Canadian coastal ports	69	69	0	Exchanging ballast water	Treating ballast water prior to releasing in Canadian waters	Installation of BWMS, and ongoing operation of system (treatment). They will no longer exchange ballast water.
Vessels operating exclusively in the Canada-U.S. Great Lakes	92	18	74	Not required to manage ballast water	Treating ballast water prior to releasing in Canadian waters	Installation of BWMS, and ongoing operation of system (treatment)
Vessels operating exclusively in Canadian waters subject to performance standards	275	50	225	Not required to manage ballast water	Treating ballast water prior to releasing in Canadian waters	Installation of BWMS, and ongoing operation of system (treatment)
Vessels operating exclusively in Canadian waters subject to equivalent compliance regime	5 960	5 960	0	Not required to manage ballast water	Equivalent compliance regime prior to releasing in Canadian waters	Incremental actions are negligible

Total vessels	6 411	6 112	299	N/A	N/A	N/A
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Costs

The costs of the Regulations will be incurred by two groups: private vessel owners and the Government of Canada. The total cost over the 23-year analytical timeframe is \$280.47 million.

Cost to private vessel owners

The costs to vessel owners is a combination of acquisition, installation, and operational costs. Vessel owners will also incur costs for developing an approved ballast water management plan, training crew members to operate BWMS, keeping records of ballast water operations, and for performing vessel surveys and certifications. The total cost to private vessel owners is estimated to be \$271.11 million.

The estimated acquisition, installation, and operational costs per vessel were extracted from two different studies (described below). Costs for the approval of the ballast water management plan, record keeping and certification activities were derived from consultation with stakeholders and with TC's subject matter experts.

Acquisition, installation and operational costs

The costs for each vessel varies depending on its characteristics (e.g. length), the type of treatment system each vessel installs, the vessel's ballast water capacity, and the level of effort associated with the installation and operation of the system. The time necessary for the installation of a BWMS will vary depending on the owner's preference and the vessel operations at the time of installation. Owners may choose to install a BWMS while the vessel is in operations (i.e. having the installation crew on board during trips). Some owners could install the system while drydocking (i.e. for surveys or certification purposes, or during vessel repairs).

The costs of acquisition, installation and operation have been grouped into four categories, based on vessel type and length.

i. Costs for self-propelled vessels of 50 metres or longer

The estimated costs for vessels that are 50 metres or longer were derived from a study commissioned by TC. ⁹ This study examined the technical feasibility and costs of acquiring, installing and operating a range of BWMS for vessels that could potentially be affected by the Regulations. Four different types of BWMS ¹⁰ were examined for the following vessel types: self-unloader with accommodation aft, self-unloader with accommodation forward and aft, tanker, straight-decker, geared bulk carrier, and gearless bulk carrier.

The study derived a three-scenario case, consisting of low, high and most probable costs for each vessel type. The most probable value is based on cost quotations from manufacturers and commercial research, while the low and high estimates account for the potential variability in final outcomes. These costs were used as a proxy for each of the impacted Canadian vessels that are 50 metres or longer. Costs were adapted based on a comparison between the impacted vessel characteristics and the characteristics of the studied vessel types. For example, a vessel registered in Canada as a ferry, that is 50 metres or longer and 5 000 gross tonnage or more, is assumed to incur the costs of a "bulker-geared" vessel in the STX study. Table 2 presents the most probable costs for the least expensive option of BWMS, by vessel type.

ii. Costs for self-propelled vessels shorter than 50 metres that will meet the Convention's performance standard

The cost estimates for vessels shorter than 50 metres were derived from a study prepared for Isle Royale National Park (Houghton, MI). ¹¹ This study examined seven existing BWMS, based on their compatibility with smaller vessels and their ability to meet the performance standards described in the Convention. Among these systems, only one of them was available and compatible with the Convention standards (the Hyde Guardian). The costs associated with installing the Hyde Guardian system on the vessel "Ranger III" were used as a proxy for average acquisition, installation, and operation costs for each impacted vessel shorter than 50 metres.

iii. Costs for non-self-propelled vessels that will meet the Convention's performance standard

Due to the lack of information on the ballast water piping and electrical systems of non-self-propelled vessels, the compliance costs for this group can vary significantly from the results of this analysis. It is likely that they will adopt a case-by-case approach to comply, and the exact costs for each case cannot be estimated by TC.

Following TC's subject matter expert opinion, the cost to these vessels is assumed not to exceed that of installing the Hyde Guardian on the vessel "Ranger III." This analysis used this cost as a proxy for the acquisition, installation, and operational costs to non-self-propelled vessels with a gross tonnage equal or over 3 000.

Table 2: Acquisition, installation, and operational cost per vessel, by vessel type (least expensive BWMS, undiscounted)

Vessel type	Acquisition and installation costs	Annual operation costs
Self-unloader with accommodation aft	\$3,152,874	\$208,814
Self-unloader with accommodation forward and aft	\$2,692,831	\$137,034
Tanker	\$848,307	\$16,314
Straight decker	\$2,399,186	\$141,384
Bulker-gearred	\$858,095	\$20,664
Bulker-non-gearred	\$2,836,390	\$95,706
Vessel shorter than 50 metres, and non-self-propelled vessels	\$330,622	\$3,489

Considering the expected number of affected vessels, and the previously mentioned average costs for each vessel category, the present value total costs of acquisition, installation, and operation were estimated. Table 3 summarizes the most probable costs for all affected privately owned Canadian vessels.

Table 3: Total installation and operational costs to privately owned Canadian vessels (in millions)

Vessel type	Affected vessels	Cheapest BWMS	Acquisition and installation costs	Operation costs	Total costs
Self-unloader with accommodation aft	19	Chemical injection	\$41.36	\$24.58	\$65.94
Self-unloader with accommodation forward and aft	8	Chemical injection	\$12.56	\$5.18	\$17.74
Tanker	16	UV	\$9.15	\$1.35	\$10.50
Straight decker	10	Chemical	\$16.79	\$8.94	\$25.72

		injection			
Bulker-gearred	29	UV	\$18.76	\$2.97	\$21.73
Bulker-non-gearred	8	UV	\$14.05	\$3.58	\$17.64
Vessel shorter than 50 metres and non-self-propelled	349	Hyde Guardian	\$78.45	\$6.61	\$85.06
Total	439	N/A	\$191.12	\$53.20	\$244.33

Given that all treatment options meet the requirements of the Regulations, it is assumed that vessel owners will choose the lowest cost option. As a result, the total installation and operation costs to private vessel owners is \$244.33 million. Other possible options are further examined in the sensitivity analysis.

iv. Costs for vessels permitted to comply through the equivalent compliance regime

There are almost 6 000 vessels that fall under the equivalent compliance category. However, their characteristics and operational purposes vary significantly, and so do the procedures that vessel owners would adopt to comply with the Regulations. While data are not available to estimate their compliance cost, based on the opinion of TC's subject matter experts, vessel owners are only expected to make minor operational adjustments to comply. For this reason, it is not expected that vessel owners would incur significant cost.

Training costs

Vessel crew will require training to operate a BWMS. Initial training may be delivered by the system vendor as part of their delivery service to the vessel. Vessel owners will need to develop training materials and train new crew members.

Training costs are estimated to be \$22,500 per vessel for initial training on delivering a BWMS (material preparation, trainers, staff time, and travel, as needed). Updates and refreshers will be part of ongoing training for crews and are estimated to be an additional \$3,500 per vessel, annually. The total training costs for vessel owners are estimated to be \$15.32 million.

Survey costs

All impacted vessels will be subject to an initial survey when the vessel's BWMS is put in service. Ballast water management certification is to be endorsed following a satisfactory annual survey to maintain the validity of the certification. The certification will be reissued every five years. Based on industry consultation, it is estimated that the initial issuance and each re-issuance of the certification (every five years) will cost \$2,075 per vessel, and the annual endorsement of the survey will result in \$2,075 per vessel. The total cost for this requirement is estimated at \$7.31 million.

Ballast water management plan, and record book costs

Canadian vessels will need to develop and implement an approved ballast water management plan. This plan must comply with the Convention provisions, and include matters relating to safety, ballast water operations, coordination procedures, and crew responsibilities. TC delegated approval of the ballast water management plan to recognized organizations in fall of 2017. Each vessel will be required to have an initial management plan and provide it to recognized organizations to make sure that it meets standards. The cost related to the initial development of the management plan will be \$1,740 per vessel. Reviews may take between two and eight weeks, and will cost approximately \$4,000 (for up to two reviews). The total cost associated with developing an approved ballast water management plan is \$1.61 million.

As part of the management plan, all impacted vessels will be required to have a record book on board, exclusively for ballast water operations, in a prescribed format. Based on industry consultation, it is estimated that this requirement will cost approximately \$950 per vessel, annually. The total cost associated with the record book is \$2.54 million.

Costs to government

The costs to government are expected to be \$9.37 million over the 23-year analytical timeframe. This figure includes BWMS installation and operational costs for government-owned vessels, and costs related to the enforcement of the Regulations.

Costs for government-owned vessels

There are 12 government-owned vessels that will be affected by the Regulations (5 owned by the federal government and 7 owned by provincial governments). Government vessels that will be required to comply are those engaged in commercial activities. The government costs associated with the installation, and operation of BWMS, and other compliance activities related to ballast water management (training, management plan and record book, surveys and certification) are estimated to be \$3.91 million.

Costs for compliance promotion, enforcement and regulatory administration

The costs associated with the development, promotion, and awareness of the Regulations, and the general administration of the ballast water program is \$5.45 million.

There will be no incremental costs to the Government resulting from Port State Control inspections because the existing inspection program under the *Canada Shipping Act, 2001* is not specific to any regulation. Ballast water inspections are included within the existing Port State Control regime and will be subject to prioritization with the other regulatory areas, within the overall resources available for Port State Control.

Benefits

The benefits of the Regulations are presented as monetized and qualitative benefits. The monetized benefits represent how a reduction in the number of non-native aquatic species introduced to Canadian waters will impact Canadians. The total monetized benefit over the 23-year analytical period is estimated to be \$981.85 million. Qualitative benefits are presented as avoided environmental damage and avoided loss of biodiversity to the Canadian marine ecosystem.

Monetized benefits

The monetized benefits represent the cost savings that will result from reducing the number of aquatic invasive species introduced and spread within Canadian waters through ballast water release.

The monetized benefits presented in this analysis are derived from the expected reduction of invasive species introduced to Canadian waters, as a result of the Regulations, and is limited to species that would be introduced to Canadian ports through ballast waters tanks, and consequently spread to other Canadian ports. As such, the monetized benefits do not include the reduction in spread of species that were introduced to Canadian waters through other vectors.

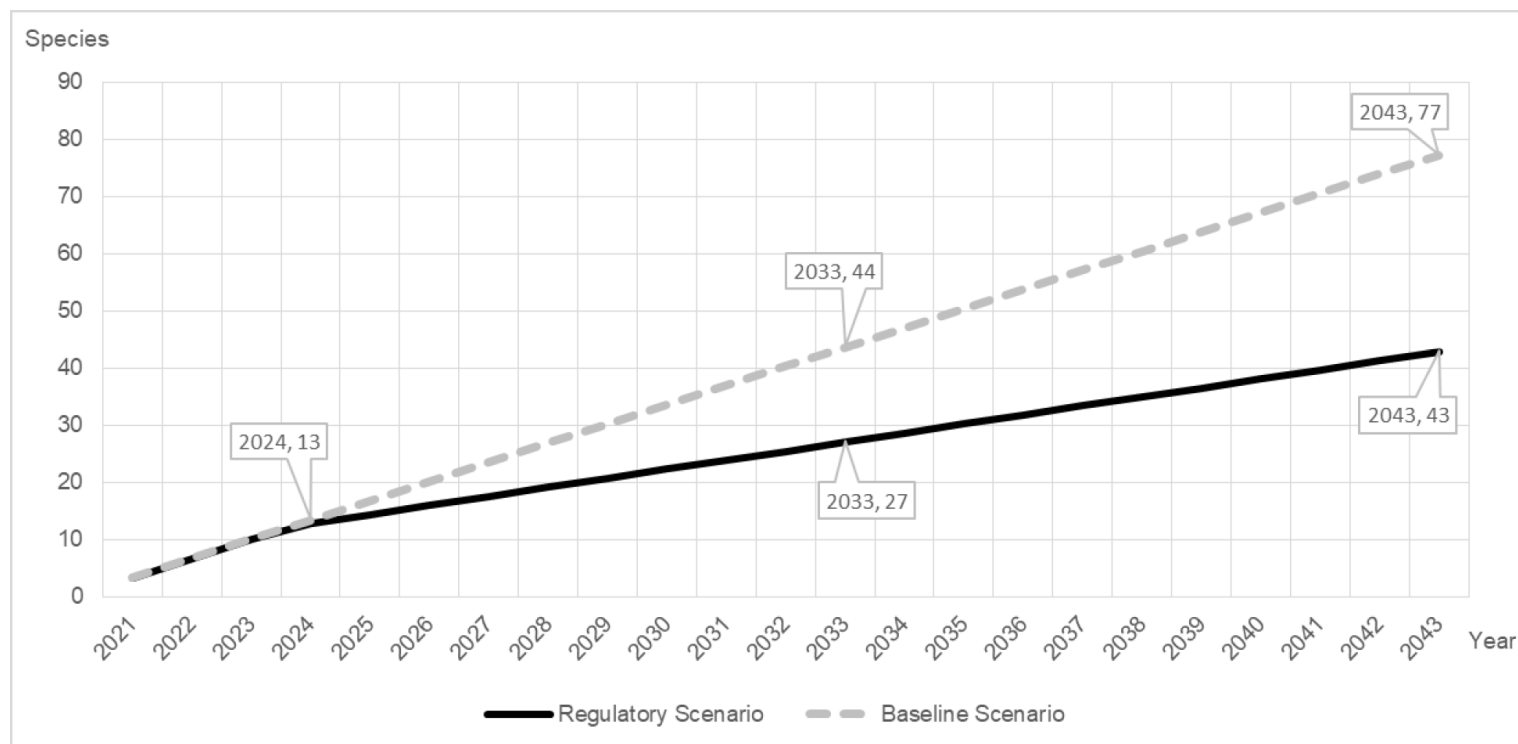
For the purpose of monetizing the benefits of the Regulations, this analysis uses the zebra mussel invasion in the Ontario Great Lakes as a proxy for the national level costs of an aquatic invasive species with severe impacts. The estimated costs of the zebra mussel invasion in the Ontario Great Lakes are representative of a regional-cost estimate, and are considered to be a conservative measure of the costs of an aquatic invasive species at the national level. The details on how this value was used to estimate the total monetized benefits are described in the “Total cost savings” section below.

Reduction of non-native species introduction

Given the various requirements and compliance dates as previously described, this analysis uses a multi-stage approach to assess the effectiveness of the Regulations on reducing the number of non-native species introduced to Canadian waters. ¹² The Regulations are expected to decrease the introduction and spread of non-native aquatic species in Canada, resulting in cost savings.

In the baseline scenario, approximately 3 non-native aquatic species would be introduced to Canada per year, totalling around 77 non-native species within the 23-year analytical period. In the regulatory scenario, assuming that ballast water treatment is effective on 50% of the transits, the number of species decreases to around 2 per year, with a total of 43 in the same period. ¹³ Therefore, the Regulations will result in a total reduction of 34 non-native species being introduced to Canadian waters, where approximately 5 of them are expected to cause severe economic or environmental damage. ¹⁴ Figure 1 shows the cumulative number of species introduced to Canadian waters in the baseline and regulatory scenarios over the 23-year analytical timeframe.

Figure 1: Number of species per year in the baseline and regulatory scenarios



► Figure 1 - Text version

Source: Fisheries and Oceans Canada (DFO)

Reduction of spread

DFO studies (Drake et al. 2020; ¹⁵ DFO 2020) indicate that international transits (using ballast water exchange) can be responsible for up to 13 zooplankton establishments per year (nationwide) while in-scope domestic vessels (no management) facilitate an estimated 48 zooplankton establishments per year within the Great Lakes region; thus, it is evident that secondary spread represents a significant source of non-native species establishments with consequent economic and environmental impacts.

The costs of an invasion are expected to be proportionate to its spread. By requiring domestic vessels to manage their ballast water prior to release, the Regulations are expected to reduce the level of spread significantly, and consequently their costs. To estimate the cost savings resulting from reduced spread of invasive species, this analysis compares establishment rates from domestic and Canada-United States Great Lakes vessel transits. These rates are extracted from the DFO study (2020) under domestic shipping scenarios when treatment is effective on 50% of transits. The baseline spread rate is represented by the “no management” scenario, and the regulatory spread rates are extracted from the scenarios where the domestic and United States–Canada Great Lakes fleet gradually adopt treatment, as to reflect the gradual compliance schedule of the Regulations.

Cost savings per expected avoided aquatic invasive species

The costs that would result from a species that has yet to invade an ecosystem cannot be known. To estimate the Regulations' potential cost savings, the impact of species that have already invaded an environment was used as a proxy for the costs of future invasions. The zebra mussel, for example, is an aquatic invasive species that has had significant consequences on a number of Canadian industries and residents, especially those that rely on water supply for their daily activities. Studies that have estimated the impacts of zebra mussels in the Canadian context are used as the main source of possible cost-avoidance values as a result of the Regulations.

Marbek (2010) studied the costs of mitigating zebra mussel invasion in the Ontario Great Lakes region, to determine how an aquatic invasive species could impact different Canadian industries. Since the cost reduction estimated in this study is found to be detailed and specific to Canadian context, it is used as one of the inputs in the cost-benefit analysis estimates for the Regulations, by using a benefit transfer methodology.¹⁶

Based on the experience of mitigating the introduction and spread of zebra mussels in the Ontario Great Lakes during a 20-year period, an annual average cost saving per aquatic invasive species was calculated. These cost savings represent the annual mitigation costs on both residents and industries in Ontario that depend on the Great Lakes for their water supply. The annual average cost of mitigating zebra mussels in the Ontario Great Lakes was estimated to be \$245.19 million (undiscounted). This estimate is representative of the mitigation costs at a regional level, and is therefore considered to be a conservative measure of the costs of an aquatic invasive species at the national level.

Severity curve of aquatic invasive species

To estimate the severity of an aquatic invasive species' introduction, a study on the recruitment patterns of zebra mussels was used as the main source of information.¹⁷ This study monitored the density of the zebra mussel population in a small Eastern Ontario river. Such patterns are taken as a proxy for the expected severity of an invasion with significant economic and environmental impact.

Total cost savings

Using the expected annual cost saving, the expected reduction in domestic rate of spread,¹⁸ and the species' expected severity levels, this analysis estimates how preventing the invasion of aquatic species with severe economic and environmental impacts will benefit Canadians in the next 23 years.

Based on an analysis from the United States Coast Guard in 2012,¹⁹ on average, the cost of a non-native aquatic species with severe economic or environmental impact would be 49% of that associated with zebra mussel invasions. This percentage is applied to the estimated annual costs that resulted from the zebra mussel invasion, as described above. The severity rate, which is based on the level of maturity and population density of a species,²⁰ is then applied to the

expected annual cost savings of an invasion. As a result, the estimated annual cost savings would reach its peak the year when the population of the species is expected to reach its maximum level. The expected average annual costs of a single non-native species with severe impacts at its maximum maturity level are estimated to be \$120.69 million (undiscounted).

Table 4 presents the annual cost savings as a result of the Regulations. The total cost savings are estimated to be \$981.85 million over the 23-year analytical timeframe.

Table 4: Present value cost savings, by year (in millions)

Year	Cost savings
2021	\$0.00
2022	\$0.00
2023	\$0.00
2024	\$0.00
2025	\$0.00
2026	\$0.00
2027	\$0.10
2028	\$2.51
2029	\$6.85
2030	\$12.21
2031	\$47.80
2032	\$57.37
2033	\$65.48
2034	\$70.06
2035	\$73.77
2036	\$76.69
2037	\$78.91
2038	\$80.51
2039	\$81.57
2040	\$82.14
2041	\$82.29
2042	\$82.07
2043	\$81.52
Total	\$981.85

Qualitative benefits

Aquatic invasive species pose a risk to native species, which can result in loss of biodiversity and, in some cases, extinction, as they represent direct competition for food and space. Many fish, such as the blue pike and the Lake Ontario Atlantic salmon, have declined significantly in population, or even disappeared from the Great Lakes due to non-indigenous organisms.

Zebra mussels, for example, feed on plankton and compete directly with small fish, resulting in fewer food sources for larger fish. Only eight years after the introduction of zebra mussels in the Great Lakes, over 20 mollusk species were extinct. Another aquatic invasive species that could have been introduced and spread via ballast water release is the ruffe, a freshwater fish commonly found in Europe and Asia. Its introduction has been one of the sources of disruption in Lake Superior's ecosystem, as they are able to reproduce quickly and compete with other species (like the yellow perch) for food supply.

In addition, untreated ballast water release can also carry and spread non-native invasive species that were introduced through vectors other than those introduced through international transit releasing at Canadian ports. These could include species that were introduced to ports in the United States and later spread to Canadian waters, as well as species that were not originally introduced to Canadian waters through ballast water (e.g. water gardens, aquarium trade). Due to a lack of data, this analysis does not quantify the benefits from reducing the spread of species that were not introduced through ballast water release in Canadian ports; however, it is important to note that the Regulations will decelerate or delay the spread and establishment of non-native aquatic species, regardless of how they were introduced to Canadian waters. Moreover, this analysis does not monetize the benefits from reducing the spread of species that were introduced to Canadian waters prior to the coming into force of the Regulations.

Cost-benefit statement

Number of years: 23 years (2021 to 2043)
Base year for costing: 2017
Present value base year: 2021
Discount rate: 7%

Table 5: Monetized costs (in millions)

Impacted stakeholder	Description of cost	Coming into force year (2024)	Other relevant year (2030)	Final year (2043)	Total (present value)	Annualized value
Industry	Installation, acquisition and operation of BWMS	\$31.45	\$68.21	\$9.02	\$244.33	\$21.68
	Training, surveys and certification, and management plan	\$4.91	\$10.43	\$2.87	\$26.78	\$2.38
Government	Compliance costs	\$0.47	\$1.47	\$0.13	\$3.91	\$0.35
	Compliance promotion and enforcement	\$0.48	\$0.48	\$0.48	\$5.45	\$0.48
All stakeholders	Total costs	\$37.30	\$80.59	\$12.49	\$280.47	\$24.88

Table 6: Monetized benefits (in millions)

Impacted stakeholder	Description of benefit	Coming into force year (2024)	Other relevant year (2030)	Final year (2043)	Total (present value)	Annualized value
Canadians	Cost-savings from avoided aquatic invasive species	\$0.00	\$22.44	\$361.19	\$981.85	\$87.10
All stakeholders	Total benefits	\$0.00	\$22.44	\$361.19	\$981.85	\$87.10

Table 7: Summary of monetized costs and benefits (in millions)

Impacts	Coming into force year (2024)	Other relevant year (2030)	Final year (2043)	Total (present value)	Annualized value
Total costs	\$37.30	\$80.59	\$12.49	\$280.47	\$24.88
Total benefits	\$0.00	\$22.44	\$361.19	\$981.85	\$87.10
NET IMPACT	-\$37.30	-\$58.15	\$348.69	\$701.38	\$62.22

Sensitivity analysis

A sensitivity analysis was conducted to test uncertainties around key variables. Two different approaches are used to analyze uncertainty in the cost-benefit estimates: a single variable sensitivity analysis and scenario sensitivity analysis.

Single-variable sensitivity analysis

The single-variable sensitivity analysis examines how the estimated net benefits would change if an assumption around one variable is changed at a time. The following variables were considered for the single-variable sensitivity analysis: the cost of a BWMS, the vessel owner's decision over BWMS, the percentage of non-native aquatic species causing severe impact, and the costs of an invasive species relative to that of the zebra mussel.

In the central case, it is assumed that vessel owners would choose the least expensive BWMS. Moreover, three costs were estimated for each of the systems that were deemed compliant with the Convention's performance standards. The sensitivity analysis considers situations in which all vessel owners would choose to install the UV system, as well as the case where owners would install the most expensive BWMS. The sensitivity analysis also considers the scenario where the costs of the least expensive systems would be either higher or lower than those used in the central case. Results are presented in tables 8.1 to 8.4.

In terms of benefits, the central case assumed that the percentage of non-native aquatic species causing severe economic impacts was 15% of the total species introduced through ballast water in the analytical period. The central case also assumes that the cost of future aquatic invasive species will be 49% of that incurred by the zebra mussel invasion. Sensitivity of results were analyzed by changing these two assumptions (see tables 8.1 to 8.4 below).

Table 8.1: Single-variable sensitivity analysis — BWMS choice

Note: The “least expensive BWMS” net benefit and cost-benefit ratio values represent the central case.

Variable	Net benefit (in millions)	Cost-benefit ratio

Least expensive BWMS	\$701.38	3.50
All vessels install UV systems	\$668.74	3.14
Most expensive BWMS	\$579.52	2.44

Table 8.2: Single-variable sensitivity analysis — cost estimate for the least expensive BWMS

Variable	Net benefit (in millions)	Cost-benefit ratio
Low-cost estimate	\$736.15	4.00
High-cost estimate	\$641.94	2.89

Table 8.3: Single-variable sensitivity analysis — percentage of non-native species causing severe impact

Variable	Net benefit (in millions)	Cost-benefit ratio
18%	\$897.75	4.20
10%	\$374.10	2.33

Table 8.4: Single-variable sensitivity analysis — cost of an invasive species, relative to the zebra mussel

Variable	Net benefit (in millions)	Cost-benefit ratio
34%	\$406.36	2.45
56%	\$1,194.82	5.26

Scenario sensitivity analysis

A scenario sensitivity analysis was conducted where lowest, most probable, and highest benefits and costs were considered. For this exercise, costs and benefits are evaluated at their extreme values simultaneously such that the net benefits can be analyzed at their upper and lower bounds. The variables that were considered for this analysis are the same used for the single-variable sensitivity analysis, but they are combined simultaneously, arriving at six different scenarios, as presented in Table 9.

Table 9: Net benefits under scenario analysis (in millions)

		Benefits		
		Lowest	Most probable	Highest
Costs	Lowest	\$212.18	\$736.15	\$1,524.64
	Most probable	\$177.42	\$701.38	\$1,489.88
	Highest	\$55.55	\$579.52	\$1,368.01

Distributional analysis

The Regulations will affect the Pacific, Ontario, and Atlantic regions the most. Approximately, 36% of the impacted vessels are registered in the Pacific region, 30% in the region of Ontario, 20% in the Atlantic region, and 13% in the Quebec Region. In terms of costs, vessels registered in Ontario represent 40% of the total costs, vessels registered in the Quebec

region represent 22%, and those registered in the Pacific region represent 21%.

Table 10: Costs to Canadian flagged vessels, by region

Region	Number of vessels	Total cost (in millions)
Atlantic	101	\$42.17
Ontario	103	\$110.95
Pacific	179	\$57.32
Quebec	62	\$3.27
The North	6	\$61.30
Total	451	\$275.02

Of the 451 vessels affected by the Regulations, 12 are owned by government entities, while the rest are privately owned. Table 11 shows a breakdown of the costs to privately and government-owned vessels.

Table 11: Number of affected vessels and costs by ownership type

Ownership	Number of vessels	Total cost (in millions)
Private	439	\$271.11
Government	12	\$3.91
Total	451	\$275.0255

Small business lens

The small business lens applies, as there are impacts on small businesses associated with the Regulations.

Based on available data, it was determined that a total of 172 businesses will be impacted by the Regulations. At least 34 of these businesses were found to be small businesses. These 34 businesses represent a total of 38 affected vessels, and the total cost for these vessels is estimated to be \$12.67 million in the analytical timeframe. It is important to note, however, that it was not possible to determine business size for 45 of the affected businesses, and it would be reasonable to assume that a portion of them might be small businesses.

In order to minimize the burden on all businesses, a transition period from the publishing of the Regulations and the coming-into-force date for compliance with the standard was included.

In the case of small and non-self-propelled vessels that operate domestically, TC has provided flexibility options, as permitted by the Convention, to establish a best-effort equivalent compliance regime (the costs associated with implementing these requirements are negligible). While the size or propulsion type of a vessel cannot be used as an indicator of the size of the business that owns or operates it, this flexibility is expected to cover the most part of domestic vessels operated by small businesses, reducing their compliance costs significantly. It is expected that small vessels travelling internationally (that do not qualify under equivalent compliance) are owned by large businesses.

Table 12: Compliance costs to small businesses (in millions)

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Activity	Annualized value	Present value
Installation, and operation of BWMS	\$0.94	\$10.60
Training, surveys and certification, and management plan	\$0.17	\$1.88
Total compliance cost	\$1.11	\$12.48

Table 13: Administrative costs to small businesses (in millions)

Activity	Annualized value	Present value
Record book	\$0.02	\$0.19
Total administrative cost	\$0.02	\$0.19

Table 14: Total compliance and administrative costs to small businesses (in millions)

Totals	Annualized value	Present value
Total cost (all impacted small businesses)	\$1.12	\$12.67
Cost per impacted small business	\$0.03	\$0.37

One-for-one rule

The one-for-one rule applies, as the Regulations result in an incremental change in administrative burden on businesses. The Regulations repeal existing regulations and replace them with a new regulatory title, which results in no net increase or decrease in regulatory titles.

The Regulations will result in an incremental administrative burden on businesses due to the ballast water record book requirement. Affected vessels will be required to have a record book on board, exclusively for ballast water operations, in a prescribed format. This record book shall be made available for inspection at all reasonable times to duly authorized officials during survey and certification of the vessel conducted by a flag State inspector or an organization recognized by the flag State or during Port State Control Verification. The annualized administrative costs for all of the affected industry are estimated to be \$114,533, or \$666 per business (23 years, 2012 dollars, discounted to 2012 at a 7% discount rate).

Regulatory cooperation and alignment

These Regulations are being introduced to align Canadian regulations with the Convention, fulfilling Canada's obligations as a party to it. In so doing, it also aims to align to the extent feasible with the differing United States regimes for ballast water, which are still evolving. Notably, the United States requires that any BWMS operated in its waters be approved by the United States Coast Guard using different tests than internationally approved standards. Canadian, United States, and international vessels will need to comply with the relevant United States regimes as well as the Convention when operating in the United States, such as in the United States Great Lakes waters.

Strategic environmental assessment

In accordance with the *Cabinet Directive on the Environmental Assessment of Policy, Plan and Program Proposals*, a preliminary scan concluded that a strategic environmental assessment is not required.

Gender-based analysis plus

No gender-based analysis plus (GBA+) impacts have been identified for these Regulations, as they relate to the requirement to give effect to Canada's international obligations under the International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004.

Implementation, compliance and enforcement, and service standards

Implementation

The Regulations come into force on the day upon which they are registered. However, associated enforcement actions would not be invoked until the Regulations are formally published in the *Canada Gazette*, Part II.

These Regulations apply to vessels depending on their operations, ballast water capacity and size, and specific compliance dates are set out in section 11 of the Regulations.

Compliance and enforcement

The enforcement of the Regulations will continue under the overall established compliance and enforcement mechanisms for TC and, as such, marine safety inspectors will continue to enforce the Regulations during normal periodic inspections.

If an operator is found to be non-compliant with the Regulations, they could be required to pay a penalty which could range from \$600 to \$25,000, depending on the infraction (as outlined in the proposed consequential amendments to the *Administrative Monetary Penalties and Notices (CSA 2001) Regulations* for the *Ballast Water Regulations*). The operator could also be, for example, required to cease operation, or have their vessel detained.

TC, through the delegated recognized organizations, will carry out surveys and certifications, and TC inspectors may impose penalties or corrective action to the operators when deficiencies are found. Through these activities TC will monitor compliance by vessel operators, similar to the way compliance is ensured for existing requirements. It is up to the operator to meet the requirements, if they fail to do so, enforcement action may be taken by TC which could include fines, prosecutions, or the detention of vessels.

In addition, Canada and the United States continue to co-operate on implementation of compatible ballast water requirements in accordance with the vessel discharge provisions of the 2012 Canada–United States Great Lakes Water Quality Agreement. This includes annual meetings between responsible authorities.

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Footnotes

- ^a S.C. 2019, c. 1, s. 141
- ^b S.C. 2018, c. 27, s. 692
- ^c S.C. 2018, c. 27, s. 707
- ^d S.C. 2014, c. 29, s. 75(1)
- ^e S.C. 2018, c. 27, s. 709
- ^f S.C. 2001, c. 26
- ^g S.C. 2019, c. 14, s. 31 and par. 50(e)
- ^h R.S., c. F-14
- ¹ SOR/2008-97; SOR/2012-246, s. 1
- ² SOR/2015-121
- ³ SOR/2011-237
- ⁴ For the purposes of this Regulatory Impact Analysis Statement, a vessel that could be part of more than one group has been assigned to the last such group in the order listed here.
- ⁵ The official science report has since been published: DFO (2020). Additional Analyses of Ballast Water Management Scenarios to Reduce the Establishment of Harmful Aquatic Species Across Canada and the Great Lakes (PDF). DFO Can. Sci. Advis. Sec. Sci. Resp. 2020/053.
- ⁶ United States Coast Guard (2012). *Final Rule - Regulatory Analysis and Final Regulatory Flexibility - Standards for Living Organisms in Ships' Ballast Water Released in U.S. Waters*. Washington, DC: U.S. Department of Homeland Security.
- ⁷ DFO. 2019. The risks of spreading aquatic invasive species to Canada by moving unmanaged ballast water from Canada to the U.S. within the Great Lakes region (PDF). DFO Can. Sci. Advis. Sec. Sci. Resp. 2019/030.
- ⁸ In this context, vessels that operate internationally include vessels that operate in the U.S.-Canada coasts, but not vessels that operate only in the U.S.-Canada Great Lakes and St. Lawrence River. See "Vessels operating exclusively in the U.S. and Canada Great Lakes and St. Lawrence River" section for details on compliance dates for such vessels.

- 9 STX Canada Marine (2015). Assessing the Feasibility Study of Ballast Water Treatment System Installation and Operation by Existing Vessels in the Great Lakes and St. Lawrence. (p. 87), available at Transport Canada upon request.
- 10 The four studied systems were side-stream electro-chlorination, ultraviolet sterilization, ozone injection, and chemical injection.
- 11 Isle Royale National Park (2011). Ballast Water Treatment System Evaluation for Small Vessels (PDF)
- 12 DFO (2020). Additional Analyses of Ballast Water Management Scenarios to Reduce the Establishment of Harmful Aquatic Species Across Canada and the Great Lakes (PDF). DFO Can. Sci. Advis. Sec. Sci. Resp. 2020/053.
- 13 These values include introduction by international ship transits destined for Canadian ports in the Pacific, Atlantic, Great Lakes and the St. Lawrence River (GLSLR), and Arctic, from foreign source ports.
- 14 This figure was determined using the conventional factor that 15% of invasive species cause severe economic or environmental damage. See: Windle, P. (1998). The OTA Report on Harmful Nonindigenous Species (PDF). In: Tennessee Exotic Pest Plant Council. USDA Forest Service. (pp. 71–77).
- 15 Drake, D. A. R., Bradie, J. N., Ogilvie, D., Casas-Monroy, O., and Bailey, S. A. 2020. Effectiveness of Ballast Water Exchange plus Treatment as a Mechanism to Reduce the Introduction and Establishment of Aquatic Invasive Species in Canadian Ports. DFO Can. Sci. Advis. Sec. Res. Doc. 2020/003. viii + 60 p.
- 16 Marbek. (2010). Assessing the Economic Value of Protecting the Great Lakes: Invasive Species Prevention and Mitigation.
- 17 Martel, A. & Madik, J. (2018). Twenty-six years (1990–2015) of monitoring annual recruitment of the invasive zebra mussel (*Dreissena polymorpha*) in the Rideau River, a small river system in Eastern Ontario, Canada. *Canadian Journal of Zoology* 96(10), 1071–1079.
- 18 Values derived from: DFO (2020) Additional Analyses of Ballast Water Management Scenarios to Reduce the Establishment of Harmful Aquatic Species Across Canada and the Great Lakes (PDF). DFO Can. Sci. Advis. Sec. Sci. Resp. 2020/053 Domestic scenario: Canadian Ports in the GLSLR, Atlantic, and Arctic.
- 19 United States Coast Guard (2012). *Final Rule - Regulatory Analysis and Final Regulatory Flexibility - Standards for Living Organisms in Ships' Ballast Water Released in U.S. Waters*. Washington, DC: U.S. Department of Homeland Security.
- 20 Level of maturity depends on years after initial introduction.
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Attachment B: Updated USCG Type Approved Ballast Water Management System Inventory¹

BWMS ²	Manufacturer	USCG Approval No.	Filtration	Disinfection Process	Ambient Water or Feed Restrictions?		Total Residual Oxidant Dosage in Fresh Water	Corrosive ³	Hold Time in Freshwater	Maximum Flow Rate (m ³ /hour)	Approved for Installation in Hazardous Locations? (46 CFR 111.105 ⁴)	Approved for Installation on U.S.-flag Vessels? (46 CFR Sub. F & J)	Compatible with U.S.-flag Lakers? ⁵
					Temperature	Salinity							
Optimarin Ballast System (OBS)	Optimarin AS	162.060/1/5	Yes	Ultraviolet (UV)	Yes Water: -2 – 37°C	N/A	N/A	No	Yes (>2 hours)	3,000	Yes	Yes	No ^R
PureBallast 3.0 & 3.1	Alfa Laval Tumba AB	162.060/2/5	Yes	UV	No	N/A	N/A	No	Yes (> 2.5 hours)	3,000	No	No	No ^{NA,R}
OceanSaver BWMS MKIIA and MKIIB ⁶	TeamTec BWMS AS	162.060/3/6	Yes	Electrodialysis	Yes Feed: > 17°C	Yes (>20 PSU)	1.7 mg/L	Yes	Yes (> 24 hours)	7,200	MKIIA: No MKIIB: Yes	MKIIA: Yes MKIIB: Yes	No ^{NFW,C,WQ,HT}
BalClor	Sunrui Marine Environment	162.060/4/3	Yes	Electrolysis	Yes Feed: > 5°C	Yes (>15 PSU)	7.5 mg/L	Yes	Yes (Varies ⁷)	8,500	No	No	No ^{NA,NFW,C,WQ,HT}
Ecochlor (ES), EcoOne Hybrid (HS), EcoOne (OS), Ecochlor Tanker (ET), EcoOne Hybrid Tanker (HT), EcoOne Tanker (OT)	Ecochlor, Inc.	162.060/5/5	Yes ⁸	Chemical Injection	No	With filter (ES, HS, ET, HT): N/A Without filter (HS, OS, HT, OT): > 1 PSU	With filter ⁹ (ES, HS, ET, HT): 4.25 mg/L ClO ₂ Without filter (HS, OS, HT, OT): 4.6 mg/L ClO ₂	Yes	Yes (48 hours – treated water must meet 0.2 mg/L ClO ₂ MADC)	16,200	ET, HT, OT: Yes ES, HS, OS: No	Yes	No ^{C,WQ,HT}
ERMA FIRST FIT ¹⁰	ERMA FIRST ESK Engineering Solutions S.A.	162.060/6/6	Yes	Electrochlorination	Yes Feed: > -2°C	Yes (0.9 PSU)	6 mg/L	Yes	No	3,740	Yes	Yes	No ^{NFW,C,WQ,R}
ECS and ECS-A	Techcross, Inc.	162.060/7/3	No	Electrochlorination	Yes Water: 0 – 55°C Feed: 0 – 36°C	Yes (ECS: N/A ECS-A: >3 PSU)	7.5 mg/L	Yes	Yes (48 hours)	12,000	No	No	No ^{NA,NFW,C,WQ,HT}

¹ Comparisons and information provided is for BWMS use in freshwater, only.

² U.S. Coast Guard list of type-approved BWMS is maintained online at: <https://www.dco.uscg.mil/Our-Organization/Assistant-Commandant-for-Prevention-Policy-CG-5P/Commercial-Regulations-standards-CG-5PS/Marine-Safety-Center-MS-C/Ballast-Water/TACs/>

³ International Maritime Organization G8 testing requirements include corrosion but only as it relates to tank coatings in accordance with the International Paint and Printing Ink Council.

⁴ All U.S. flag hazardous location approvals are conditional based on location and equipment. Foreign flag approvals are not included.

⁵ Items that impact compatibility with a U.S.-flag laker include: not approved by USCG for installation and operation on a U.S.-flag vessel (“NA”), not approved for use in freshwater (“NFW”) without salinity augmentation, chemical compatibility with vessel construction (“C”), discharge exceeds state drinking waters limits (“WQ”), hold times in ballast tanks versus voyage times (“HT”), and low flow rates incapable of maintaining normal load/unload rates for cargo operations (“R”).

⁶ Requires feed water storage equivalent to 2 to 5 percent of total ballast water volume.

⁷ Holding time varies depending on installed filter in the BWMS. BWMS with Filtersafe, Boll, and Kanagawa filters have a holding time greater than 24 hours. BWMS with the Luoyang Sunrui filter has no holding time.

⁸ Depends on model. Models OS and OT do not have filters. Models HS and HT have filters but may be operated without the filter.

⁹ Active Substance Dosage – Chlorine Dioxide (ClO₂)

¹⁰ Requires feed water storage equivalent to 2 percent of total ballast water volume.

Attachment B: Updated USCG Type Approved Ballast Water Management System Inventory¹

BWMS ²	Manufacturer	USCG Approval No.	Filtration	Disinfection Process	Ambient Water or Feed Restrictions?		Total Residual Oxidant Dosage in Fresh Water	Corrosive ³	Hold Time in Freshwater	Maximum Flow Rate (m ³ /hour)	Approved for Installation in Hazardous Locations? (46 CFR 111.105 ⁴)	Approved for Installation on U.S.-flag Vessels? (46 CFR Sub. F & J)	Compatible with U.S.-flag Lakers? ⁵
					Temperature	Salinity							
Purimar	S&S Co., Ltd.	162.060/8/3	Yes	Electrochlorination	Yes Water: 0 – 55°C Feed: 4 – 40°C	Yes (> 10 PSU)	2.5–3 mg/L	Yes	Yes (S Series Filter: > 24 hours F Series Filter: N/A)	10,000	No	No	No NA,NFW,C,WQ,HT
BIO-SEA B, M and L models	BIO-UV Group	162.060/9/4	Yes	UV	Yes Water: -2 – 40°C	N/A	N/A	No	No	1,400	No	Depends on Model ¹¹	No ^R
Aquarius EC <i>EXPIRED</i>	Wartsila Water Systems Ltd.	162.060/10/4	Yes	Electrolysis	Yes Water: -2 – 45°C Feed: > 15°C	Yes (>15 PSU)	10 mg/L	Yes	Yes (24 hours)	4,000	No	No	No NA,NFW,C,WQ,HT,R
HiBallast	Hyundai Heavy Industries Co., Ltd.	162.060/11/3	Yes	Electrolysis	Yes Feed: > 4°C	Yes (>15 PSU)	8 mg/L	Yes	Yes (> 24 hours)	10,000	No	No	No NA,NFW,C,WQ,HT
OceanGuard BWMS	Headway Technology Group Co., Ltd.	162.060/12/3	Yes	Electrolysis	Yes Water: 0 – 40°C	Yes (>0.85 PSU)	2.0 mg/L	Yes	Yes (> 2 hours)	5,200	No	No	No NA,NFW,C,WQ
JFE BallastAce	JFE Engineering Corporation	162.060/13/4	Yes	Chemical Injection	No	N/A	Model A: 20 mg/L Model B and U-4: 2.5-14 mg/L Target: 12 mg/L	Yes	Yes (> 24 hours)	4,000	No	No	No NA,C,WQ,HT,R
GloEn-Patrol	Panasia Co., Ltd.	162.060/14/2	Yes	UV	Yes Water: -2 – 40°C	N/A	N/A	No	GloEn-Patrol: > 48 hours GloEn-Patrol 2.0: > 24 hours	6,000	No	No	No ^{NA,HT}
BALPURE BWMS	De Nora Marine Technologies, LLC	162.060/15/4	Yes	Electrolysis	Yes Water: 0 – 50°C Feed: 15 – 50°C	Yes (>18 PSU)	7 – 15 mg/L	Yes	Yes (> 24 hours)	8,570	No	No	No NA,NFW,C,WQ,HT

¹¹ BIO-SEA B model with the HYDAC filter and alternative control model, and the BIO-SEA M models are not approved for US flagged vessels. No information is presented on other B and L models. USCG confirmed via email that BIO-SEA B model with the Filtrex filter is approved for installation on a U.S. Flag vessel.

Attachment B: Updated USCG Type Approved Ballast Water Management System Inventory¹

BWMS ²	Manufacturer	USCG Approval No.	Filtration	Disinfection Process	Ambient Water or Feed Restrictions?		Total Residual Oxidant Dosage in Fresh Water	Corrosive ³	Hold Time in Freshwater	Maximum Flow Rate (m ³ /hour)	Approved for Installation in Hazardous Locations? (46 CFR 111.105 ⁴)	Approved for Installation on U.S.-flag Vessels? (46 CFR Sub. F & J)	Compatible with U.S.-flag Lakers? ⁵
					Temperature	Salinity							
inTank BWTS	SCIENCO/FAST	162.060/16/2	No	Electrolysis/Chemical Injection ¹²	Yes Feed: 0 – 35°C	>22 mS/cm conductivity	2-5 mg/L ¹³	Yes	Yes (24 hours – may increase)	N/A ¹⁴	Yes	Yes	No NFW,C,WQ,HT
CompactClean	DESMI OceanGuard A/S	162.060/17/3	Yes	UV	No	N/A	N/A	No	Yes (> 2 hours)	1,740	Yes	Yes	No ^R
Aquarius UV	Wartsila Water Systems Ltd.	162.060/18/6	Yes	UV	Yes Water: -2 – 45°C	N/A	N/A	No	Yes (> 72 hours)	1,000	Yes	Yes	No ^{HT,R}
PureBallast 3.2	Alfa Laval Tumba AB	162.060/19/5	Yes	UV	No	N/A	N/A	No	Yes (> 2.5 hours)	3,000	Yes	Yes	No ^R
Evolution	Cathelco Ltd.	162.060/20/0	Yes	UV	Yes Water: > -2°C	N/A	N/A	No	Yes (> 48 hours)	1,500	No	Yes	No ^{HT,R}
Blue Ocean Shield (BOS)	COSCO Shipping Heavy Industry Technology Co., Ltd.	162.060/21/2	Yes	UV	Yes Water: 0 – 40°C	N/A	N/A	No	Yes (> 2 hours)	3,200	No	No	No ^{NA,R}
EcoBallast	Hyundai Heavy Industries Co., Ltd.	162.060/22/0	Yes	UV	Yes Water: < 45°C	N/A	N/A	No	Yes (> 120 hours)	2,160	No	No	No ^{NA,HT,R}
HK-(E)C	Miura Co., Ltd.	162.060/23/3	Yes	UV	Yes Water: 0 – 50°C	N/A	N/A	No	Yes (> 17 hours)	900 (intake) 1,200 (discharge)	No	No	No ^{NA,HT,R}
EcoGuardian	HANLA IMS Co., Ltd.	162.060/24/1	Yes	Electrolysis	Yes Water: 0 – 40°C Feed: >10°C	Yes (>10 PSU)	6-10 mg/L	Yes	No	6,000	No	No	No NA,NFW,C,WQ
HK-S(E)	Miura Co., Ltd.	162.060/25/0	Yes	UV	Yes Ambient: < 50°C	Yes (>1 PSU)	N/A	No	Yes (> 48 hours)	900 (intake) 1,200 (discharge)	No	No	No NA,NFW,HT,R
BAWAT BWMS Mk2	BAWAT A/S	162.060/26/1	No	Heat	No	N/A	N/A	No	Yes (40-80 seconds ¹⁵)	5,000	No	No	No ^{NA,HT}

¹² System offers two options for chemical disinfection: direct injection of NaOCl or generation on board through electrolysis.
¹³ Varies with hold time. Minimum concentration-time (CT): 120 mg-hr/L.
¹⁴ Treatment is performed in ballast water tank after uptake and before discharge. Total amount of ballast water capable of being treated with chemical capacity on board is 200,000 m³ (52 million gallons).
¹⁵ The system requires a minimum residence time between 40 and 80 seconds that is a function of the pasteurization temperature. This normally equates to approximately 4 hours.

Attachment B: Updated USCG Type Approved Ballast Water Management System Inventory¹

BWMS ²	Manufacturer	USCG Approval No.	Filtration	Disinfection Process	Ambient Water or Feed Restrictions?		Total Residual Oxidant Dosage in Fresh Water	Corrosive ³	Hold Time in Freshwater	Maximum Flow Rate (m ³ /hour)	Approved for Installation in Hazardous Locations? (46 CFR 111.105 ⁴)	Approved for Installation on U.S.-flag Vessels? (46 CFR Sub. F & J)	Compatible with U.S.-flag Lakers? ⁵
					Temperature	Salinity							
GloEn-Patrol 2.0 <i>EXPIRED</i>	Panasia Co., Ltd.	162.060/27/1	Yes	UV	Yes Water: -2 – 40°C	N/A	N/A	No	Yes (> 24 hours)	6,000	Yes	Yes	No ^{HT}
NK-O3 BlueBallast II	NK Co. Ltd.	162.060/28/0	No	Ozone	No	Yes (>1 PSU)	Ozone: 2.5 g/m ³	Yes	Yes (> 24 hours)	8,000	No	No	No NA,NFW,C,HT
NK-O3 BlueBallast II Plus	NK Co. Ltd.	162.060/29/0	No	Ozone + NaBr ¹⁶	No	N/A	Ozone: 2.5 g/m ³	Yes	Yes (> 24 hours)	8,000	No	No	No NA,C,WQ,HT
Hyde GUARDIAN-US	Hyde Marine	162.060/30/0	Yes	UV	Yes Water: 0 – 40°C	N/A	N/A	No	Yes (> 24 hours)	3,000	Yes	Yes	No ^{HT,R}
HK-(E)R	Miura Co., Ltd.	162.060/31/3	Yes	UV	Yes Water: < 50°C	N/A	N/A	No	Yes (> 72 hours)	900 (intake) 1,200 (discharge)	No	No	No ^{NA,HT,R}
ECS-HYCHLOR	Techcross, Inc.	162.060/32/0	Yes	Electrochlorination	Yes Water: 0 – 55 °C Feed: > 2.5°C	Yes (8 PSU)	5 mg/L	Yes	Yes Filtersafe Filter: >3 hours Techcross Filter: >48 hours	8,000	Yes ¹⁷	Yes	No ^{NFW,C,WQ}
LUV U1	Semb-Eco Pte Ltd.	162.060/33/0	Yes	UV	Yes Water: 5 – 55°C	Yes (> 10 PSU)	N/A	No	Yes (> 48 hours)	500	No	No	No NA,NFW,HT,R
MICROFADE II	Kuraray Co., Ltd.	162.060/34/0	Yes	Chemical Injection	No	N/A	8 mg/L	Yes	Yes (> 24 hours)	2,000	No	No	No NA,C,WQ,HT,R
oneTank	oneTank, LLC	162.060/35/2	No	Chemical Injection	No	N/A	5 mg/L ¹⁸	Yes	Yes (> 24 hours)	4,000	No	Yes	No ^{C,WQ,HT}
Seascope	Elite Marine Corporation	162.060/36/0	Yes	UV	Yes Water: -2 – 40°C	N/A	N/A	No	Yes (> 48 hours)	5,000	No	No	No ^{NA,HT}
ATPS-BLUESys	Panasonic Enviro. Systems	162.060/37/0	No	Electrolysis	Yes Water: > 2°C ¹⁹	Yes (> 5 PSU)	12 mg/L	Yes	Yes (> 24 hours)	3,600	No	Yes	No NFW,C,WQ,HT,R

¹⁶ According to information from NK Co. Ltd., addition of NaBr when operating below 1 PSU.
¹⁷ Only the Techcross ECS-HYCHLOR models with the “Ex-” prefix meet the requirements of 46 CFR 111.105 and may be installed in hazardous locations to which they are certified on a U.S.-flag vessel.
¹⁸ Minimum concentration-time (CT): 120 mg-hr/L.
¹⁹ Temperature and Salinity limitations are listed for 100% TRC. Different values if operated at 60% TRC.

Attachment B: Updated USCG Type Approved Ballast Water Management System Inventory¹

BWMS ²	Manufacturer	USCG Approval No.	Filtration	Disinfection Process	Ambient Water or Feed Restrictions?		Total Residual Oxidant Dosage in Fresh Water	Corrosive ³	Hold Time in Freshwater	Maximum Flow Rate (m ³ /hour)	Approved for Installation in Hazardous Locations? (46 CFR 111.105 ⁴)	Approved for Installation on U.S.-flag Vessels? (46 CFR Sub. F & J)	Compatible with U.S.-flag Lakers? ⁵
					Temperature	Salinity							
KBAL BWMS	Knutsen Ballast Water AS	162.060/38/2	No	Pressure Drop/UV	Yes Water: -2 – 50°C	N/A	N/A	No	Yes (> 72 hours)	3,000	No	No	No ^{NA,HT,R}
PureBallast 2.0	Alfa Laval Tumba AB	162.060/39/0	Yes	UV	Yes Water: -2 – 40°C	Yes (> 1 PSU)	N/A	No	Yes (> 22 hours)	2,500	No	No	No ^{NA,NFW,HT,R}
Senza BWMS	TeamTec BWMS AS	162.060/40/2	No	Chemical Injection	N/A	N/A	15 mg/L	Yes	Yes (> 24 hours)	3,750	No	No	No ^{NA,C,WQ,HT,R}
SKF BlueSonic BWMS	SKF Marine GmbH	162.060/41/1	Yes	Ultrasonic and UV	N/A	N/A	N/A	No	Yes (> 24 hours)	1,500	No	No	No ^{NA,HT,R}
BSKY BWMS	Wuxi Brightsky Electronic Co., Ltd.	162.060/42/1	Yes	Separation/UV	Yes Water: 0 – 55°C	N/A	N/A	No	Yes (> 72 hours)	6,000	No	No	No ^{NA,HT}
SeaCURE	Evoqua Water Technologies Limited	162.060/43/2	Yes	Electrochlorination	Yes Water: 0 – 45°C Feed: 0 – 35°C	Yes (≥ 25 PSU)	8 mg/L	Yes	Yes (> 24 hours)	6,000	Yes	Yes	No ^{NFW,C,WQ,HT}
NiBallast BWMS	Jiangsu Nanji Machinery Co., Ltd.	162.060/44/1	Yes	Deoxygenation	N/A	N/A	N/A	No	Yes (> 120 hours)	4,000	No	No	No ^{NA,HT,R}
HiBallast NF	Hyundai Heavy Industries Co., Ltd.	162.060/45/1	No	Electrolysis	Yes Feed: > 4°C	Yes (> 15 PSU)	8 mg/L	Yes	Yes (Varies) ²⁰	10,000	Yes	Yes	No ^{NFW,C,WQ,HT}
CleanBallast-Ocean Barrier System (CB-OBS)	RWO GmbH	162.060/46/0	Yes	Electrochlorination	Yes Feed: 0 – 35°C	Yes (> 0.9 PSU)	6 mg/L	Yes	Yes (24 hours)	3,000	No	No	No ^{NA,NFW,C,WQ,HT,R}
NGT BWMS	Norwegian Greentech AS	162.060/47/0	Yes	UV	N/A	N/A	N/A	No	N/A	609	No	No	No ^{NA,R}
Purestream BWMS	Atlantium Technologies Ltd.	162.060/48/0	Yes	UV	N/A	N/A	N/A	No	Yes (> 24 hours)	1,500	No	Yes	No ^{HT,R}
LanghBW BWMS	Langh Tech Oy Ab	162.060/49/0	Yes	UV	N/A	N/A	N/A	No	Yes (> 24 hours)	600	No	No	No ^{NA,HT,R}

²⁰ Holding time varies depending on electrolyzer installed. Module EM-P and EM-S have a holding time > 48 hours. Module EM-H has a hold time > 24 hours.

Attachment B: Updated USCG Type Approved Ballast Water Management System Inventory¹

BWMS ²	Manufacturer	USCG Approval No.	Filtration	Disinfection Process	Ambient Water or Feed Restrictions?		Total Residual Oxidant Dosage in Fresh Water	Corrosive ³	Hold Time in Freshwater	Maximum Flow Rate (m ³ /hour)	Approved for Installation in Hazardous Locations? (46 CFR 111.105 ⁴)	Approved for Installation on U.S.-flag Vessels? (46 CFR Sub. F & J)	Compatible with U.S.-flag Lakers? ⁵
					Temperature	Salinity							
AquaStar BWMS	AQUASTAR Co., Ltd.	162.060/50/0	Yes	Electrolysis	Yes Water: 0 – 40°C	Yes (≥ 10 PSU)	9 ml/L	Yes	Yes (> 24 hours)	5,000	No	No	No NA,NFW,C,WQ, HT
Cyeco BWMS	Shanghai Cyeco Electric Technology Co., Ltd.	162.060/51/0	Yes	UV	N/A	N/A	N/A	No	Yes (≥ 24 hours)	1,600	No	No	No ^{NA,HT,R}
ARA Plus+	SAMKUN CENTURY CO., LTD	162.060/52/0	Yes	UV	-2 – 50°C	N/A	N/A	No	Yes (> 48 hours)	1,300	No	No	No ^{NA,HT,R}
EcoGuardian NF	HANLA IMS Co., Ltd.	162.060/53/0	No	Electrochlorination	> 10°C	> 10 PSU	Range: 6 - 10 mg/L Target: 8 - 9 mg/L	Yes	Yes (> 20 hours)	6,000	No	No	No NA,NFW,C,WQ, HT
ECS - HYCHLOR™ 2.0	Techcross, Inc.	162.060/54/0	No	Electrolysis	Feed: ≥ 3°C	> 10 PSU	9.5 mg/l	Yes	Yes (≥ 23 hours)	6,000	Yes	Yes	No NFW,C,WQ,HT



Attachment C – List of U.S.-flagged Lakers

A-390
Alpena
American Century
American Courage
American Integrity
American Mariner
American Spirit
Arthur M. Anderson
Ashtabula - Defiance
Bradshaw McKee - St. Marys Conquest
Burns Harbor
Calumet
Caroline McKee - Commander*
Cason J. Callaway
Cleveland Rocks
Clyde S. VanEnkevort - Erie Trader*
Dirk S. VanEnkevort - Michigan Trader*
Dorothy Ann - Pathfinder
Edgar B. Speer
Edward L. Ryerson
Edwin H. Gott
Endeavour*
Endurance
G.L. Ostrander - Integrity
Great Republic
H. Lee White
Herbert C. Jackson
Hon. James L. Oberstar
Indiana Harbor
James L Kuber
James R. Barker
John G. Munson
John J. Boland
John Sherwin
Joseph L. Block
Joyce L VanEnkevort - Great Lakes Trader
Kaye E. Barker
Lake Guardian
Lee A. Tregurtha
M 8128
Manitowoc
Mark W Barker*
Menominee
Mesabi Miner

Paul R. Tregurtha
Philip R. Clarke
Prentiss Brown - St. Marys Challenger*
Presque Isle - Presque Isle
RTC 101
Sam Laud
Samuel de Champlain - Innovation
Spartan II
Stewart J. Cort
Undaunted - Pere Marquette 41
Walter J. McCarthy Jr.
Wilfred Sykes

Notes:

*Denotes Post-2008 Vessel

Tug – Barge combinations are listed together with a hyphen “-” separating the tug and barge.